

Visibility Filtering Bias

Iván Rendo



Introduction

Twitter / X brief controversy history:

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Before Musk: popularity bias, woke censorship

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Mainly through content moderation:

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Alternatives: Mastodon (~failed), Bluesky (Twitter copycat)

Obvious question

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Does this promotion of the right wing exist?

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Does this promotion of the right wing exist?

and...

What if no-promotion? Users necessarily better-off?

What is the optimal bias? Consequences?

What is Musk optimizing for? Users? Profits? Reach?

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How?

Structural model, traffic data (usage)

Estimate preferences of users for content across platforms

Bias: if choices $\neq$ preferences would predict

Anecdotal Evidence / Claims

(pre and post-Musk)

Anecdotal Evidence / Claims

(pre and post-Musk)

News

Software 11-18-2024 at 13:47

45

103

Elon Musk changed X algorithms for self-promotion and promotion of Republicans – study



Vadym Karpus

News writer



Anecdotal Evidence / Claims

(pre and post-Musk)

Republicans take aim at the Biden White House's emails with tech platforms

Selective releases from Rep. Jim Jordan and a federal court case have become a focus for Republicans claiming censorship on social media.

f

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TikTok and X recommend pro-AfD content to non-partisan users ahead of the German elections

f

Anecdotal Evidence / Claims

(pre and post-Musk)

Republicans take aim at the Biden White

Jim Jordan Reveals 'Smoking Gun' Against Biden

Published Aug 08, 2023 at 12:18 PM EDT

Updated Aug 08, 2023 at 1:51 PM EDT

By [Andrew Stanton](#)

Weekend Staff Writer



Newsweek Is A Trust Project Member

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News Article



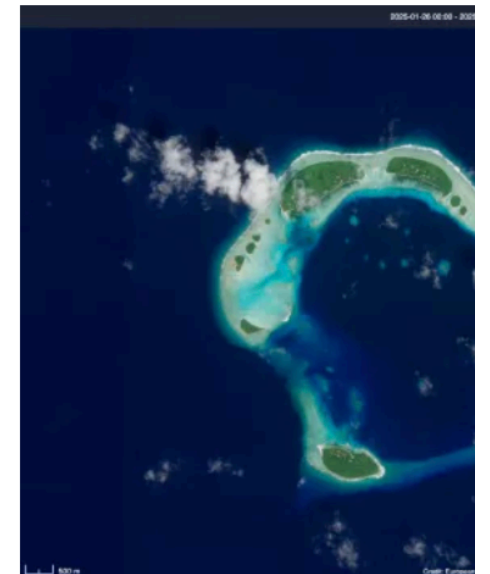
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Translate

Representative [Jim Jordan](#) unveiled his "smoking gun" evidence that the Biden administration collaborated with [Meta](#) company leaders to remove information about COVID-19.

Trending



Anecdotal Evidence / Claims

(pre and post-Musk)

Re

Jim

Published A

News Article

Representa
collaborate

The Musk Effect:

X's impact on Germany's election

Findings

The far-right AfD [outperform other German parties](#) on X. Most of this engagement from social media users is driven by Alice Weidel, the party's leader, and her [online interactions with Elon Musk](#). That has allowed the AfD to gain increased engagement on X largely from [English-language accounts](#).

The Atlantic Council's [Digital Forensic Research Lab](#) (DFRLab) and [AlgorithmWatch](#) see no evidence of either increased social media engagement of the AfD beyond the "Musk bump" to Alice Weidel, nor systematic suppression of anti-AfD accounts by X's recommender systems.

Anecdotal Evidence / Claims

(pre and post-Musk)

Re The Musk Effect:

TWITTER

Twitter suppressed stories based on requests from both Dems and GOP in 2020, but it favored liberals: Report

Twitter took 'extraordinary steps' to suppress the Hunter Biden story in 2020, journalist Matt Taibbi wrote



By **Brie Stimson** · Fox News

Published December 2, 2022 10:11pm EST

News Article

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Anecdotal Evidence / Claims

(pre and post-Musk)



POLITICO



in Ukraine Romania election Poland election | Newsletters Podcasts Poll of Polls Policy news

Twitter

De

Tw

NEWS > TECHNOLOGY

m both
report

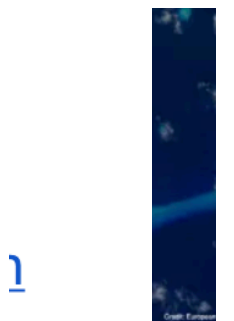
bi wrote

News Article

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Brussels will be watching whether Musk breaks EU law in far-right livestream

Up to 150 experts in Brussels and Seville will be checking whether Musk's livestream interview boosts the German far right.



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Anecdotal Evidence / Claims

(pre and post-Musk)



POLITICS



  r/Twitter • hace 2 a
hu_ssam



What's wrong with Twitter's algorithm?

 COMPLAINTS

First of all, I like the For You page on Twitter. I usually start scrolling through the Following tweets first until I finish them then go to For You page. I like exploring new accounts and following news that I am interested in but don't really want to follow people reporting it. Usually,

News Article

Representa
collaborate

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Up to 150 experts in Brussels and Seville will be checking whether Musk's livestream interview boosts the German far right.



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Anecdotal Evidence / Claims

(pre and post-Musk)



POLITICS



r/Twitter • hace 2 a

POLITICS

Latest 'Twitter Files' reveal Humorless Adam Schiff, Dems pushed to censor Biden joke

By Bruce Golding

Representa
collaborate

in far-right livestream

123 Comments

Up to 150 experts in Brussels and Seville will be checking whether Musk's livestream interview boosts the German far right.

Anecdotal Evidence / Claims

(pre and post-Musk)



POLITICS



r/Twitter • hace 2 a



INSIGHTS | LETTERS

Are algorithmic bias claims supported?

In their Special Issue Research Article, “Asymmetric ideological segregation in exposure to political news on Facebook” (28 July, p. 392), S. González-Bailón *et al.* analyzed links to news that US Facebook users shared and viewed during the 2020 election and found substantial ideological segregation in content on Facebook. They claim that

algorithm. In contrast to the conclusions of González-Bailón *et al.*, Guess *et al.* find that news feed ranking decreases, rather than increases, exposure to like-minded political content (table S20) and political news from partisan sources relative to reverse chronological feed (supplementary text S3.3).

Although evidence in both of these papers suggests that news feed ranking has limited effects on partisan content delivery, caution is warranted when drawing conclusions about the effects of ranking

the conflicting results confirm that Facebook about how algorithms shape the content

Messing also acknowledges interference for users who but minimizes the value. Focusing on this suggests users that are unintentional provide important insights relationships between and Facebook’s feed

livestream interview boosts the German far right.

Anecdotal Evidence / Claims

(pre and post-Musk)



POLITICS



The New York Times

Elon Musk Is Using X to Push His Views, and Donald Trump

Mr. Musk has become a vocal supporter of Mr. Trump. It wasn't always that way.

livestream interview boosts the German far right.

My contribution (as an economist)

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+ Users don't see everything displayed to them

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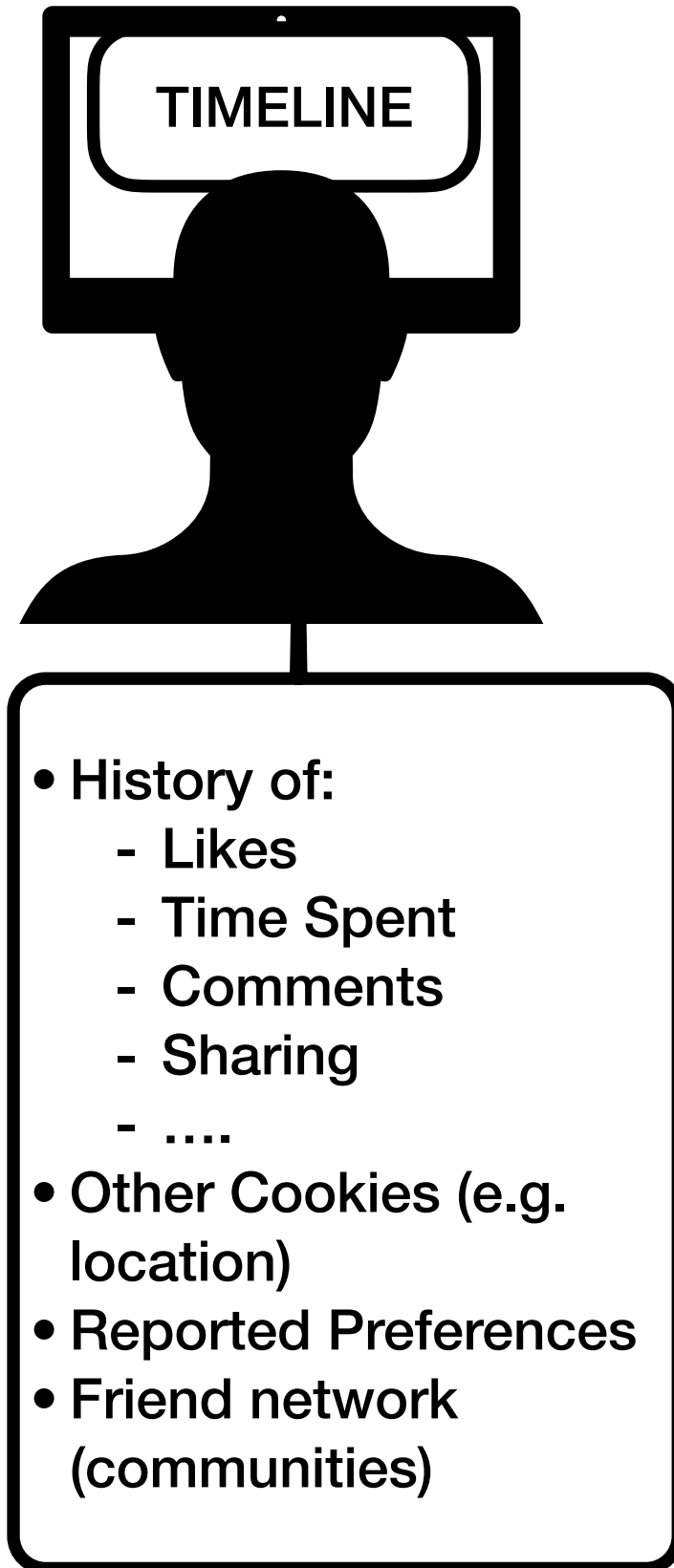
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CS does that: limited results

How X's algorithm “works”



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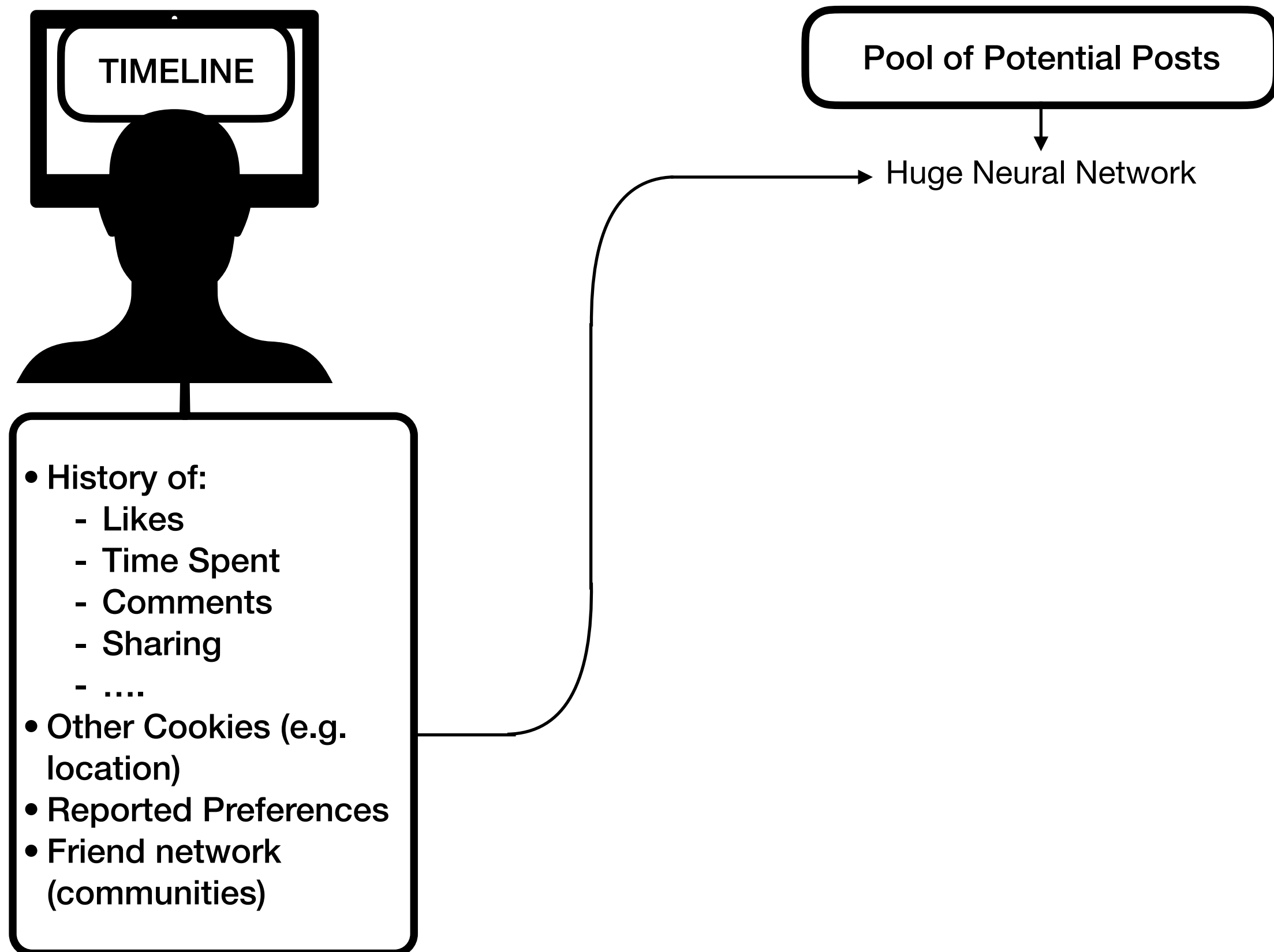
How X's algorithm “works”



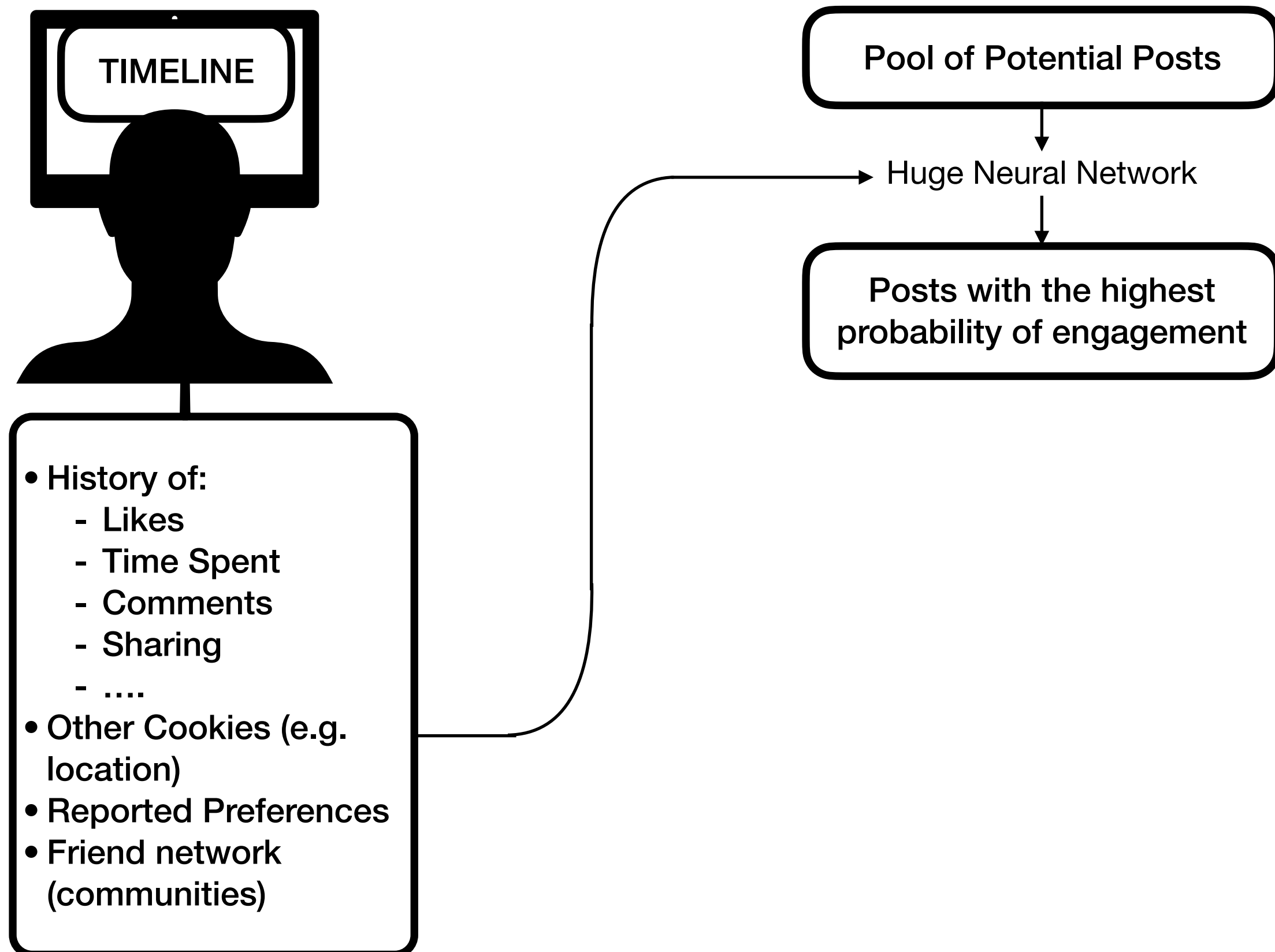
Pool of Potential Posts

- History of:
 - Likes
 - Time Spent
 - Comments
 - Sharing
 -
- Other Cookies (e.g. location)
- Reported Preferences
- Friend network (communities)

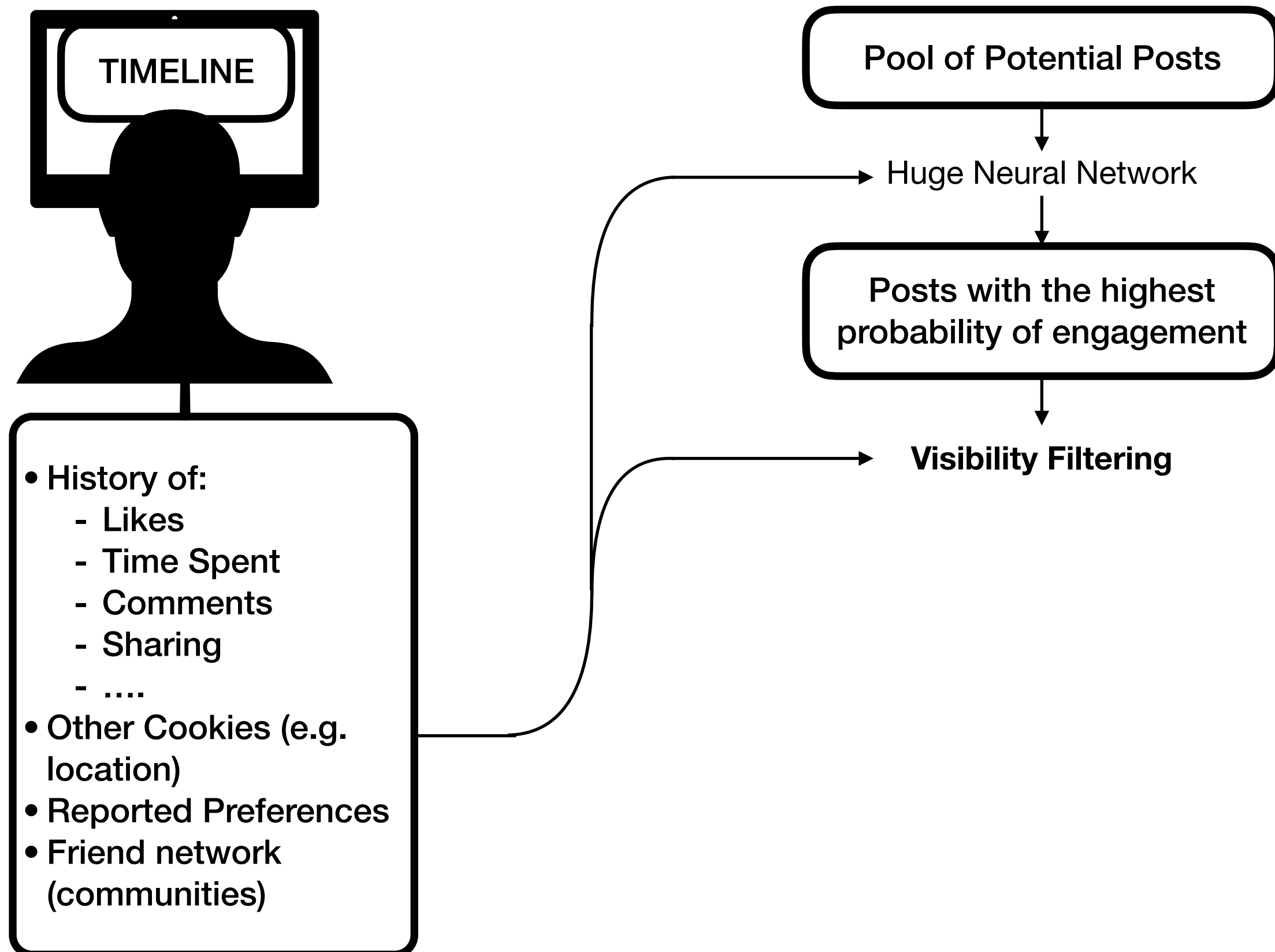
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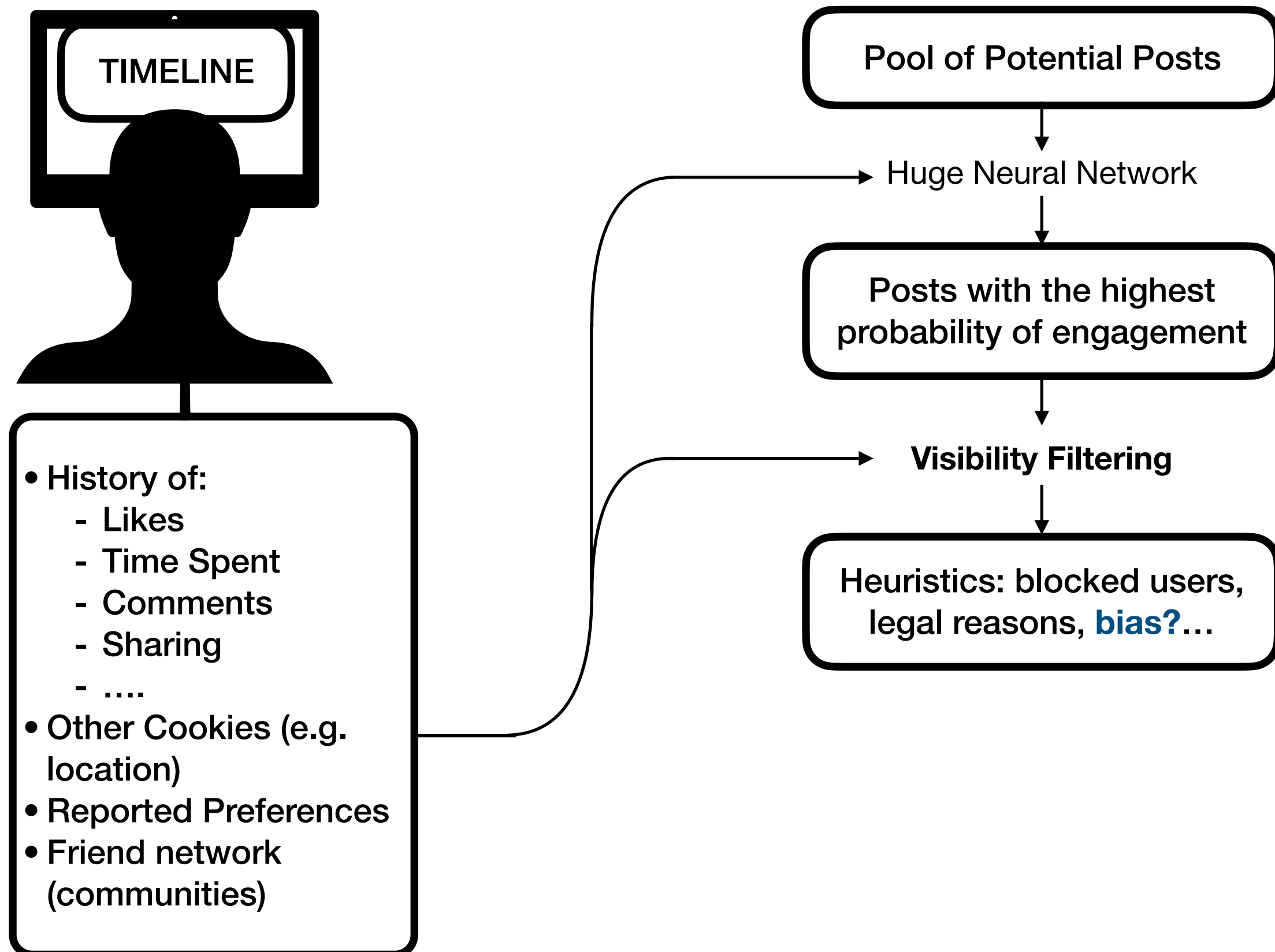
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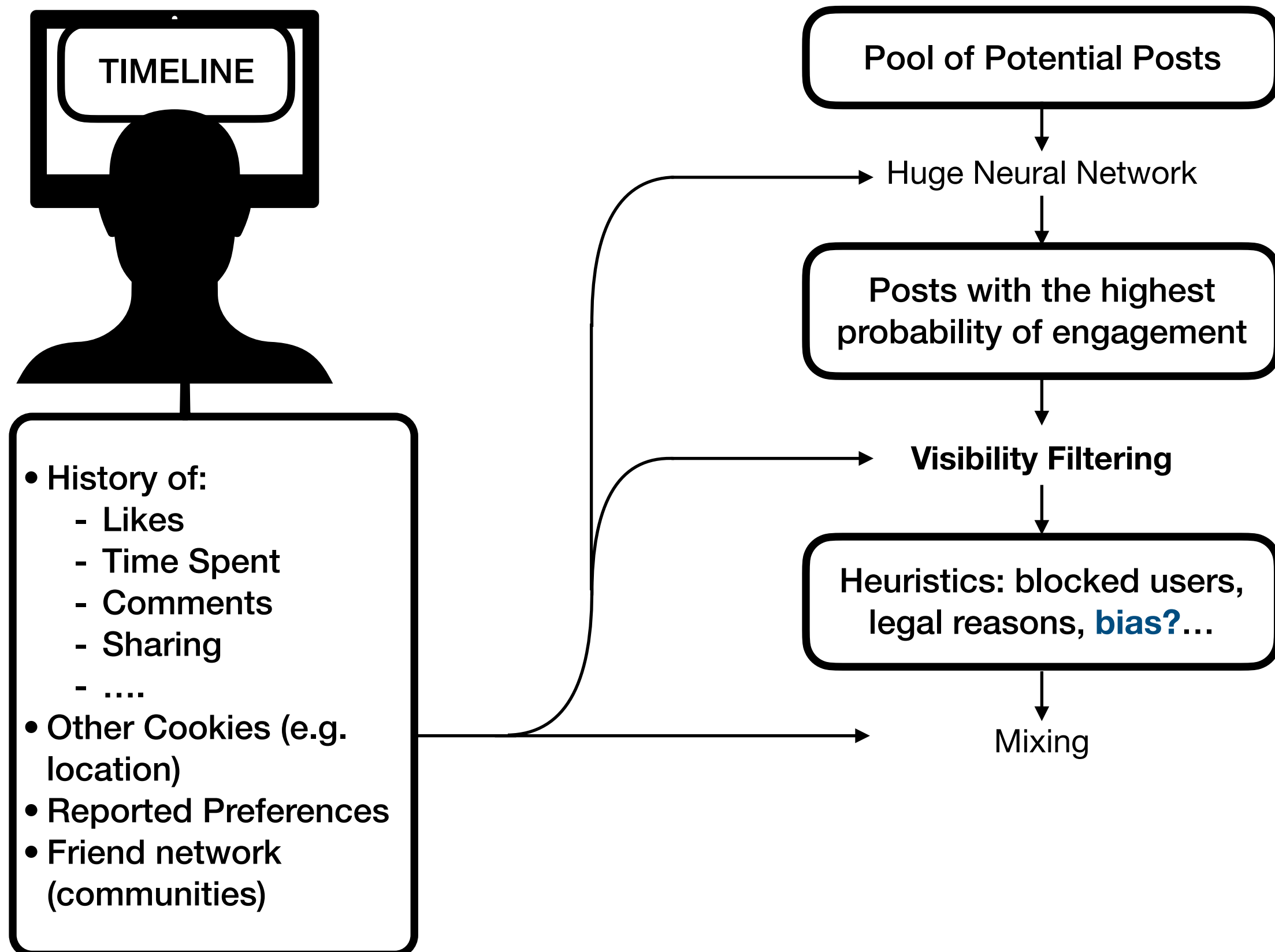
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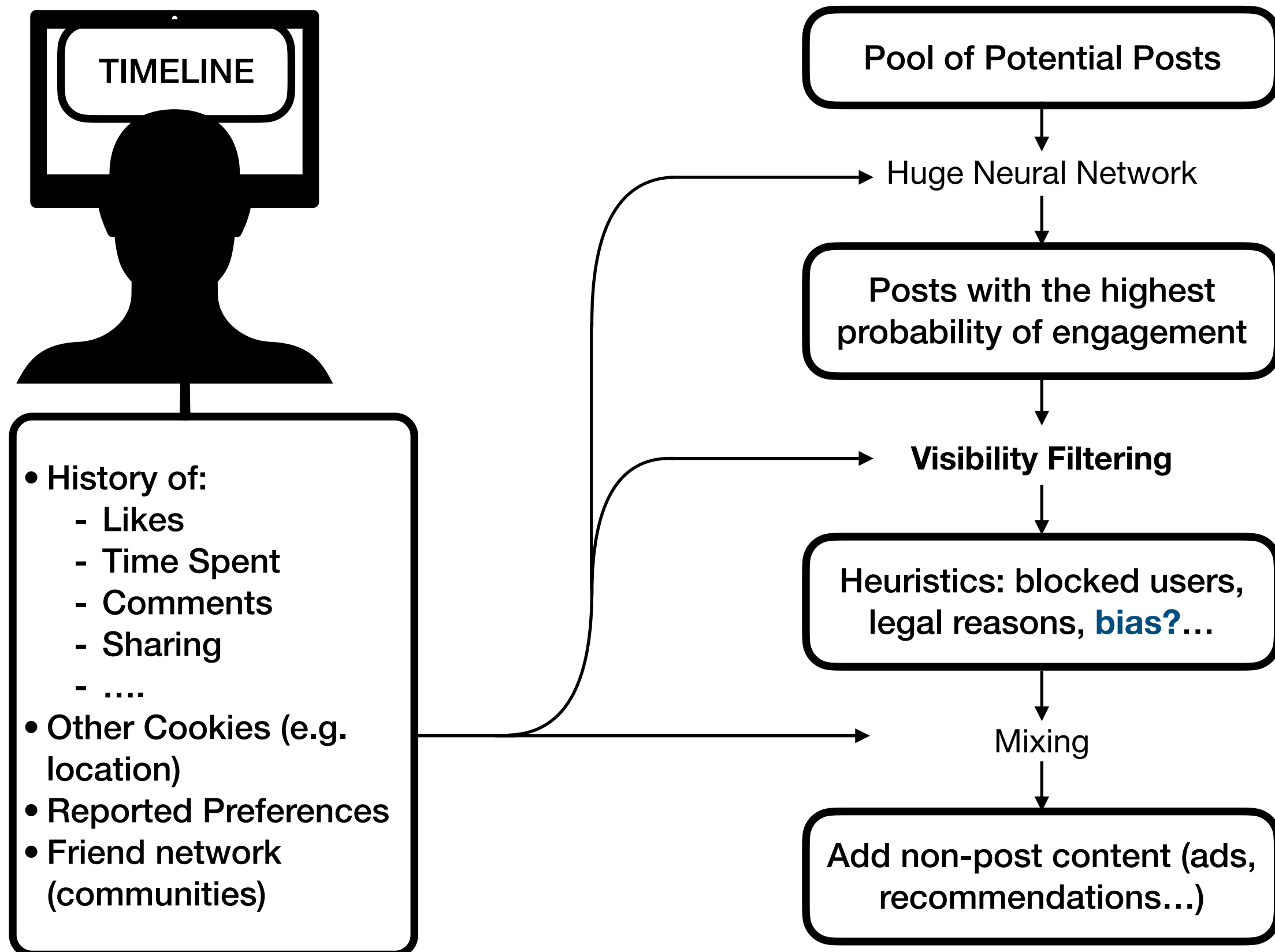
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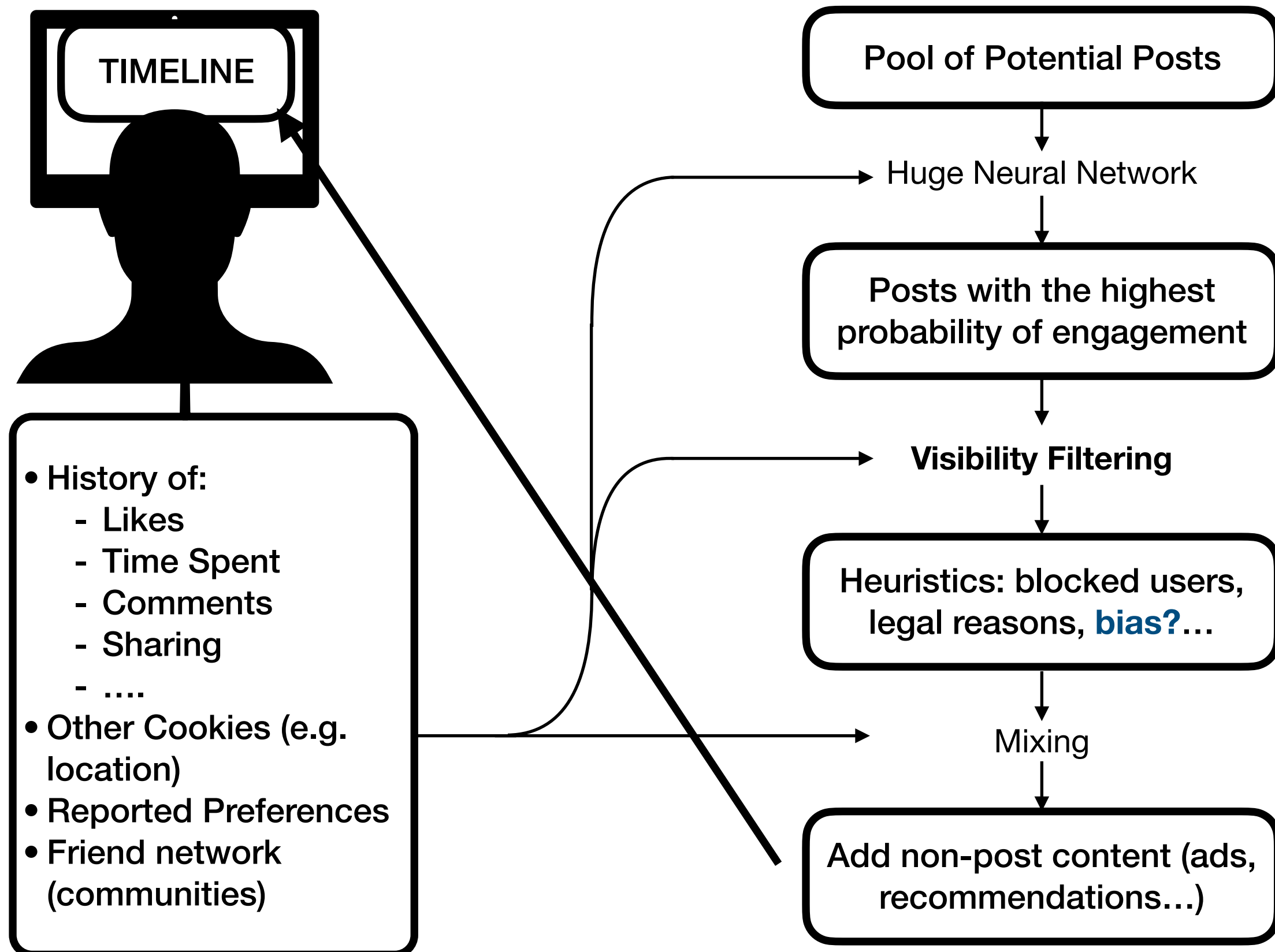
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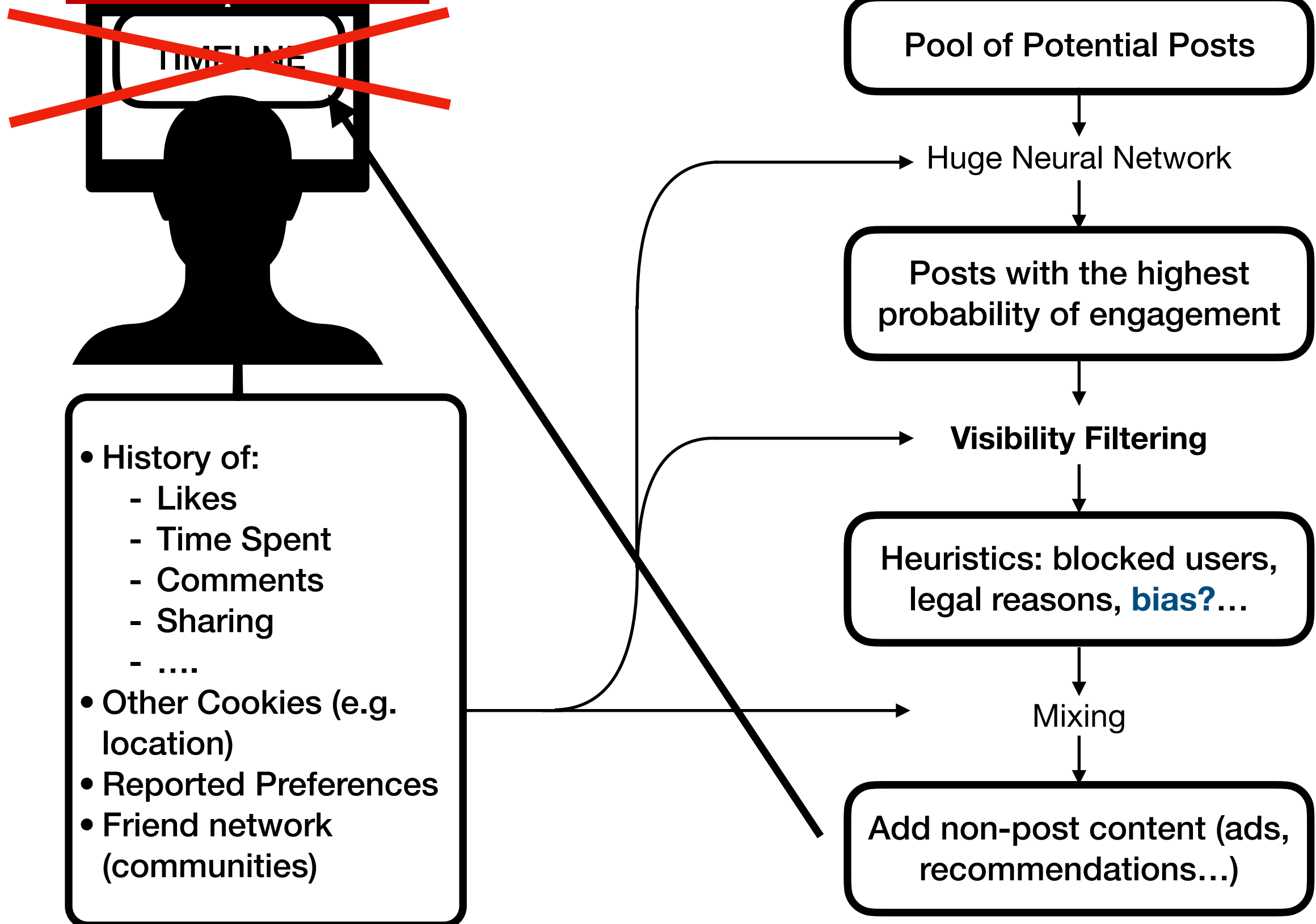


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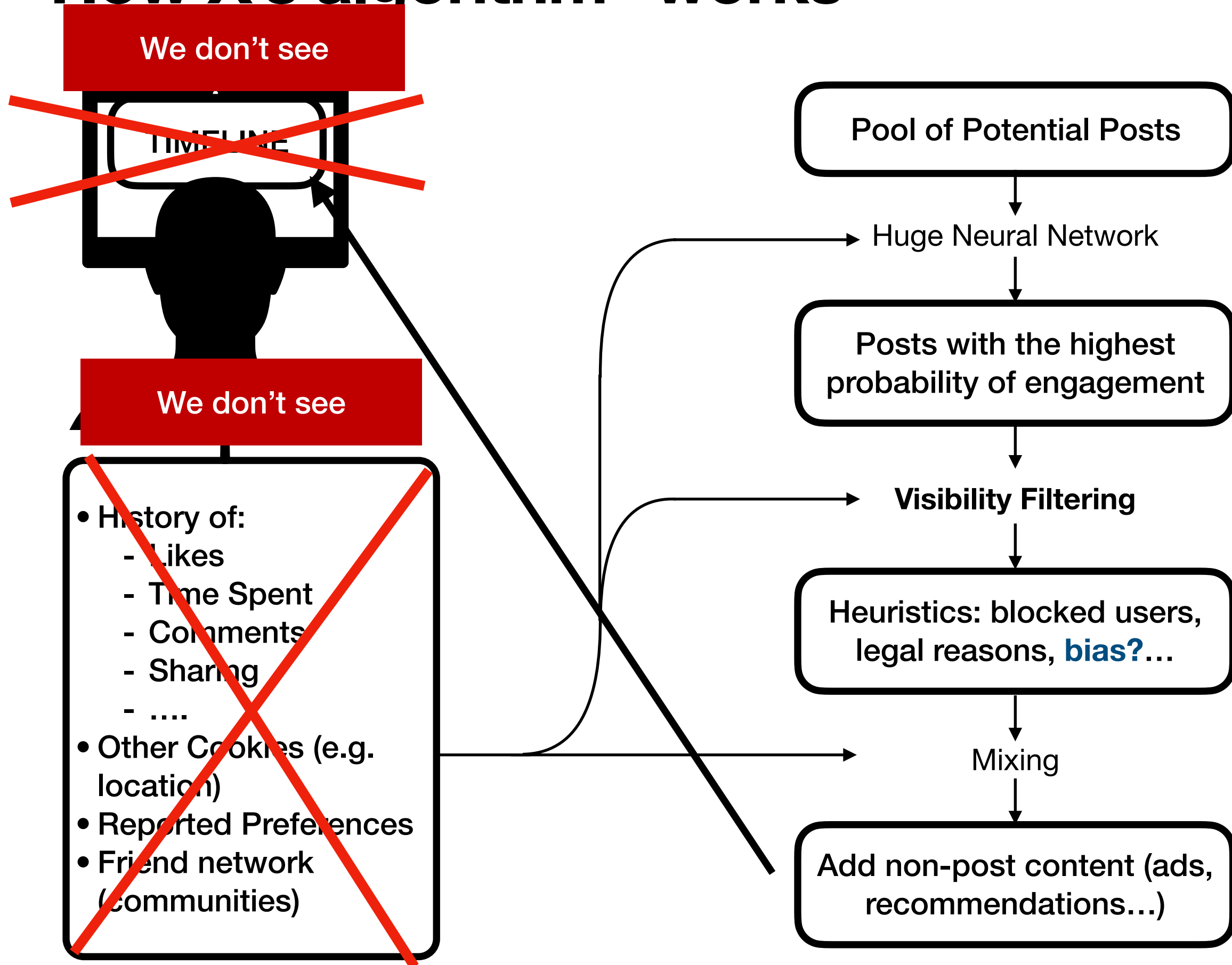


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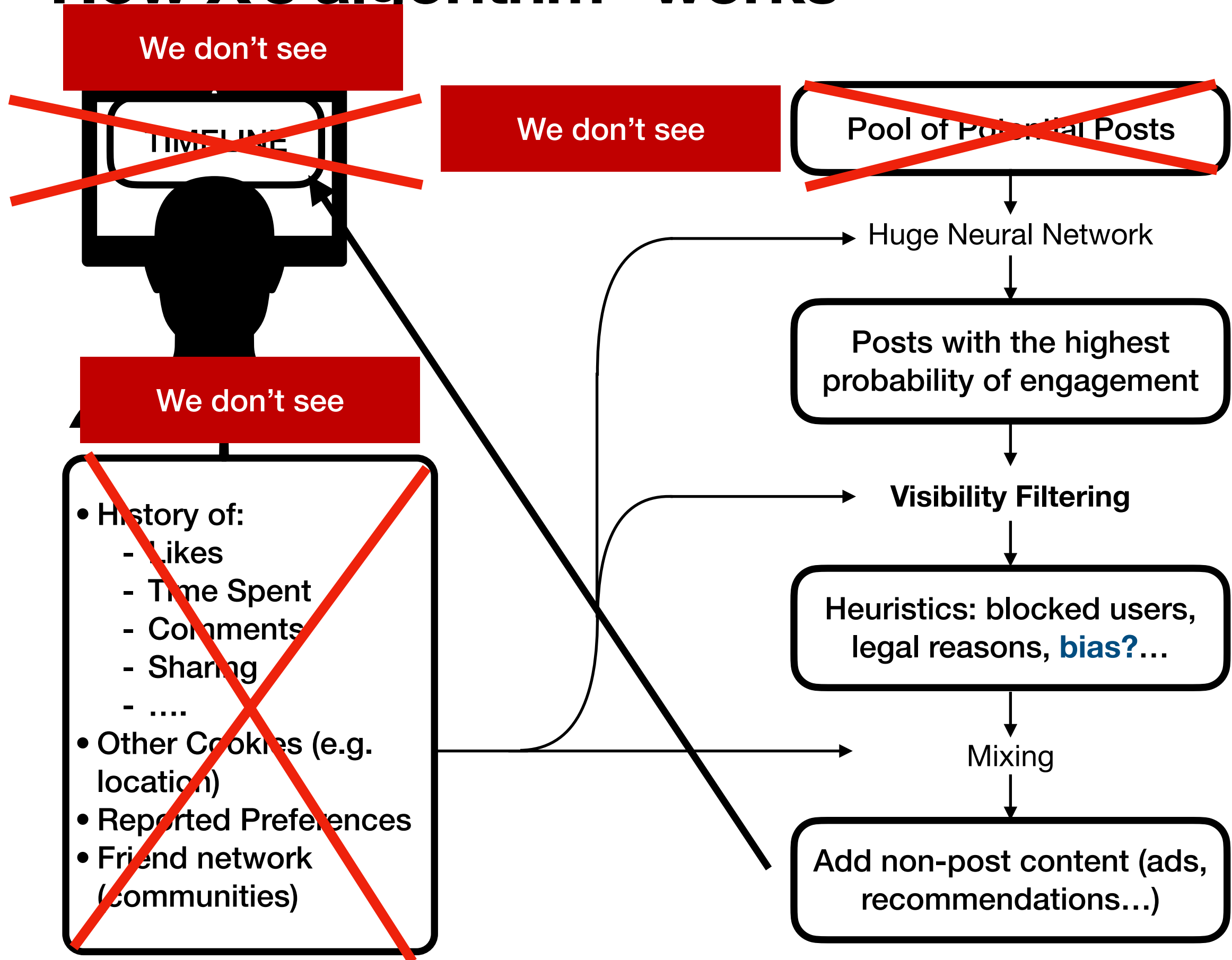
We don't see



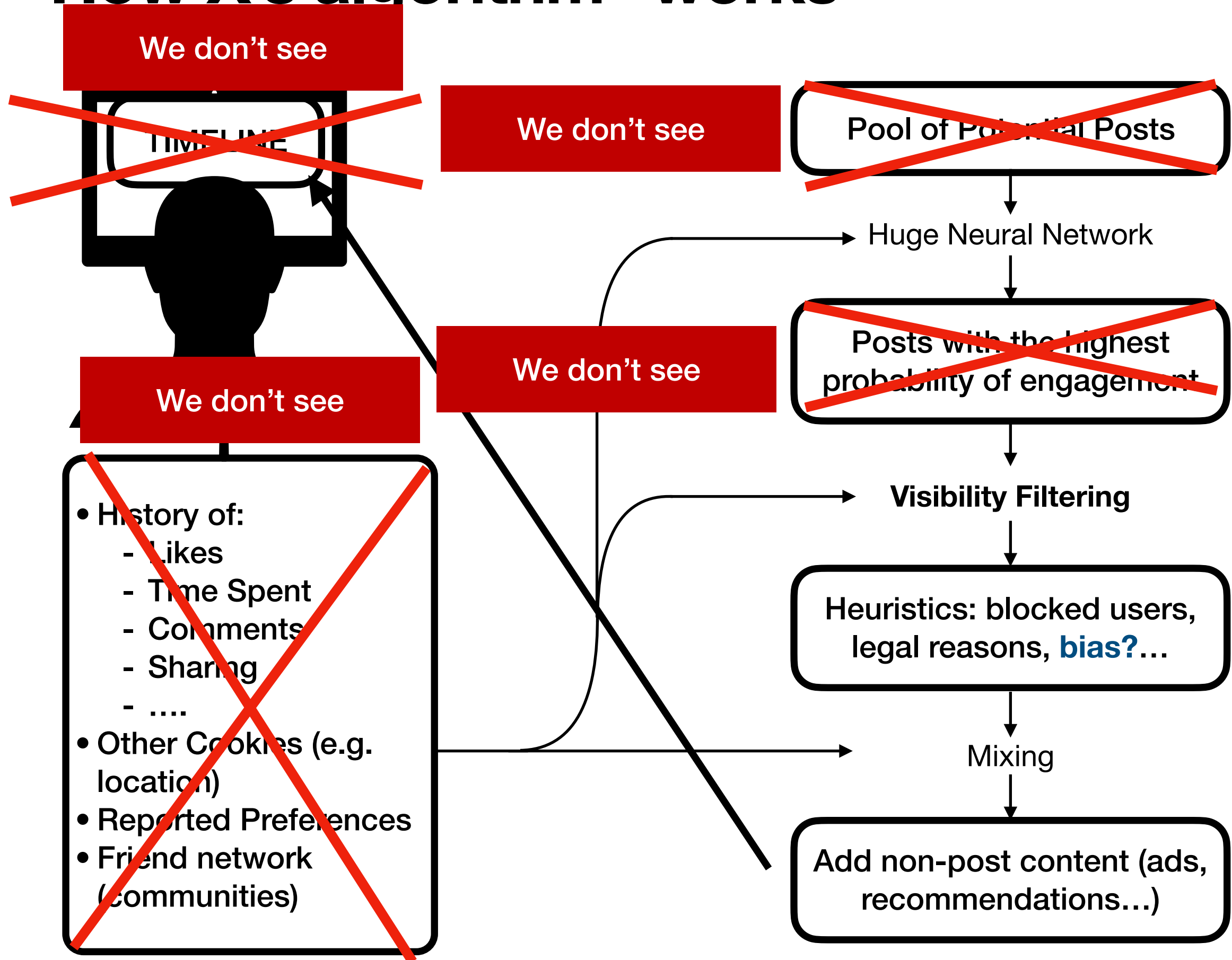
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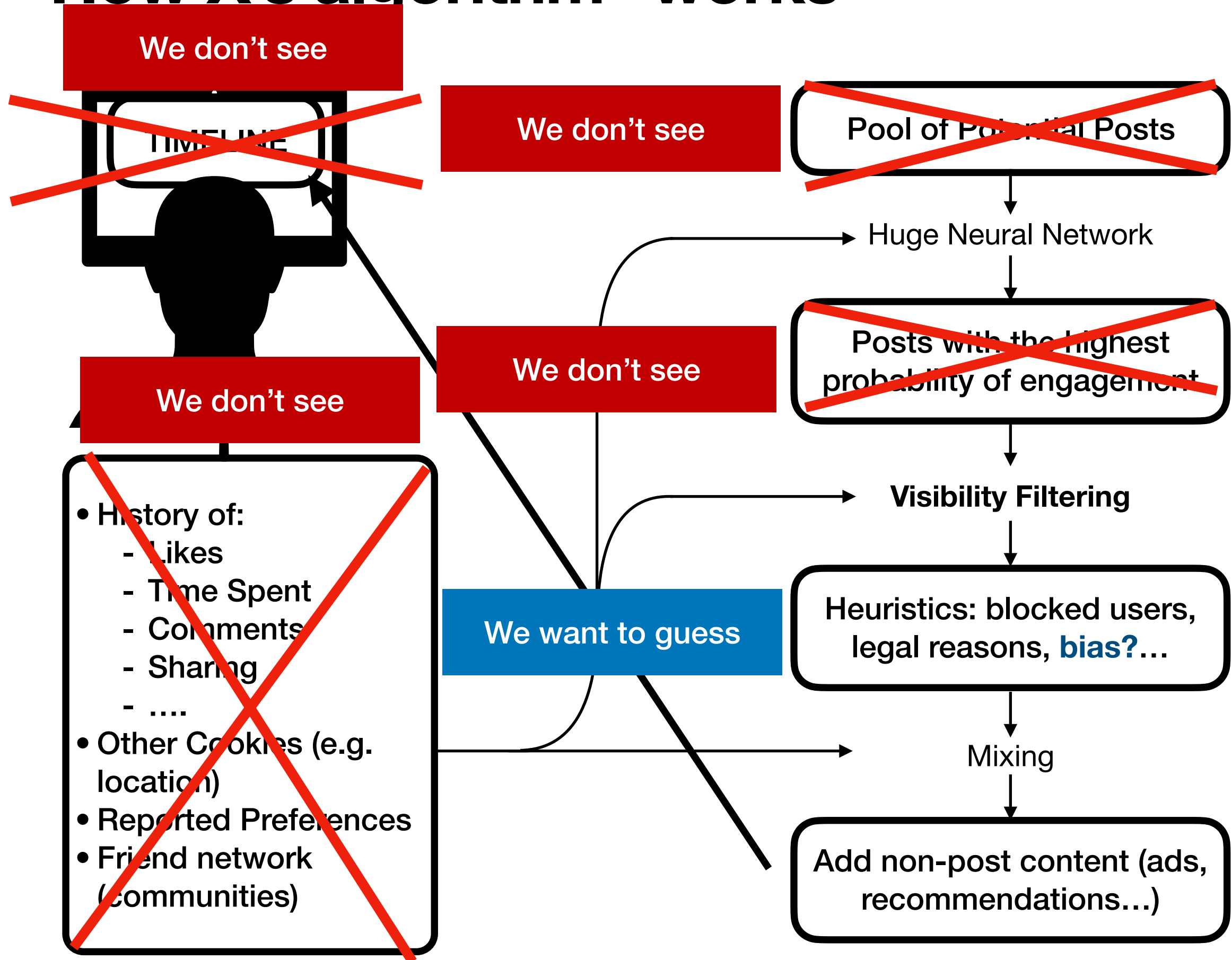
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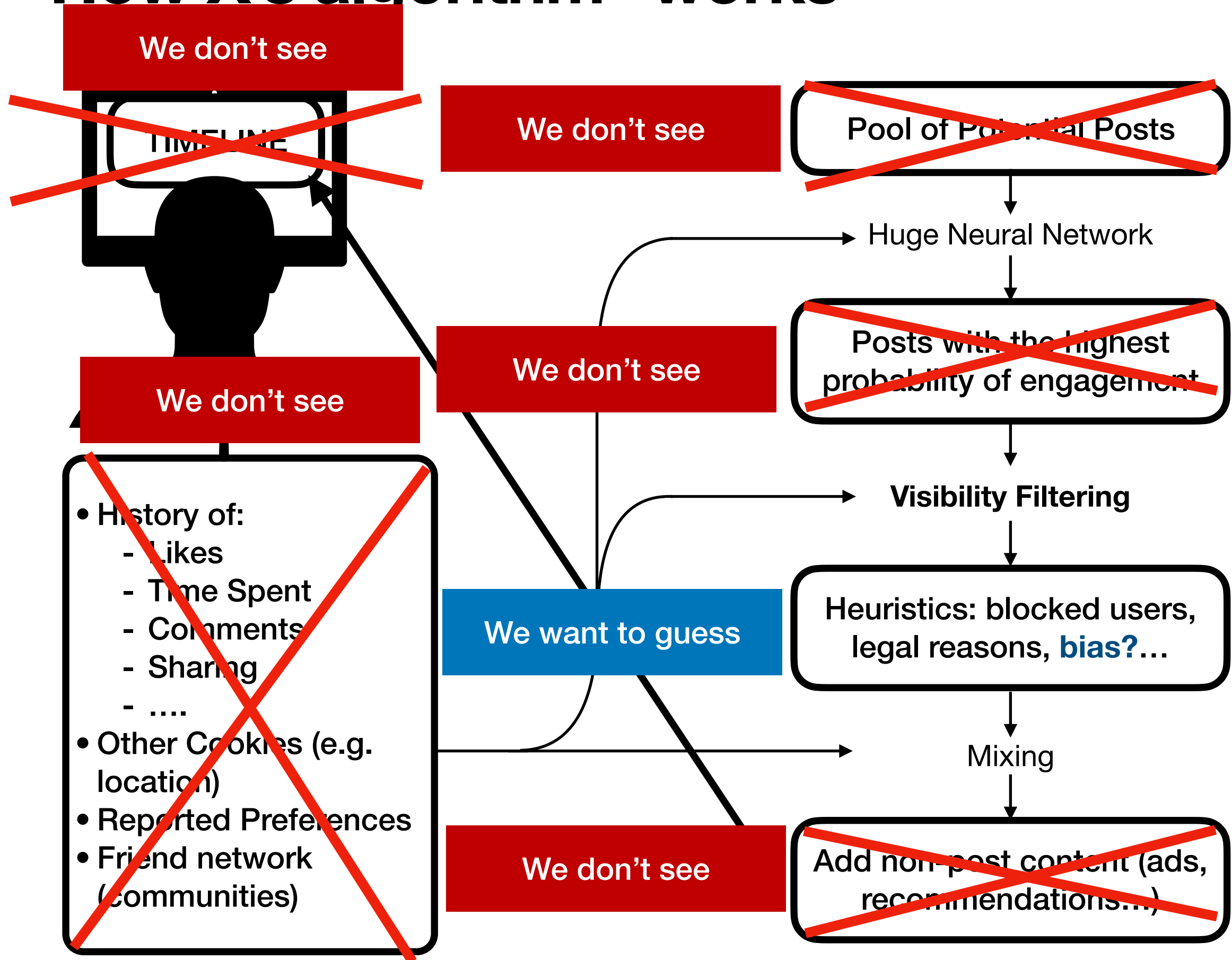
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Data

SimilarWeb traffic data (Gomez-Herrera et al., 2022; Madio et al. 2025; Carballa-Smichowski, 2025; DG-COMP vs FB)

Monthly, from 04/2022 to 04/2025

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Number of users of Le Monde* who

- Also used X/Twitter
- Also used Bluesky
- Also used Facebook
- Multihomed on several of these
- Used none

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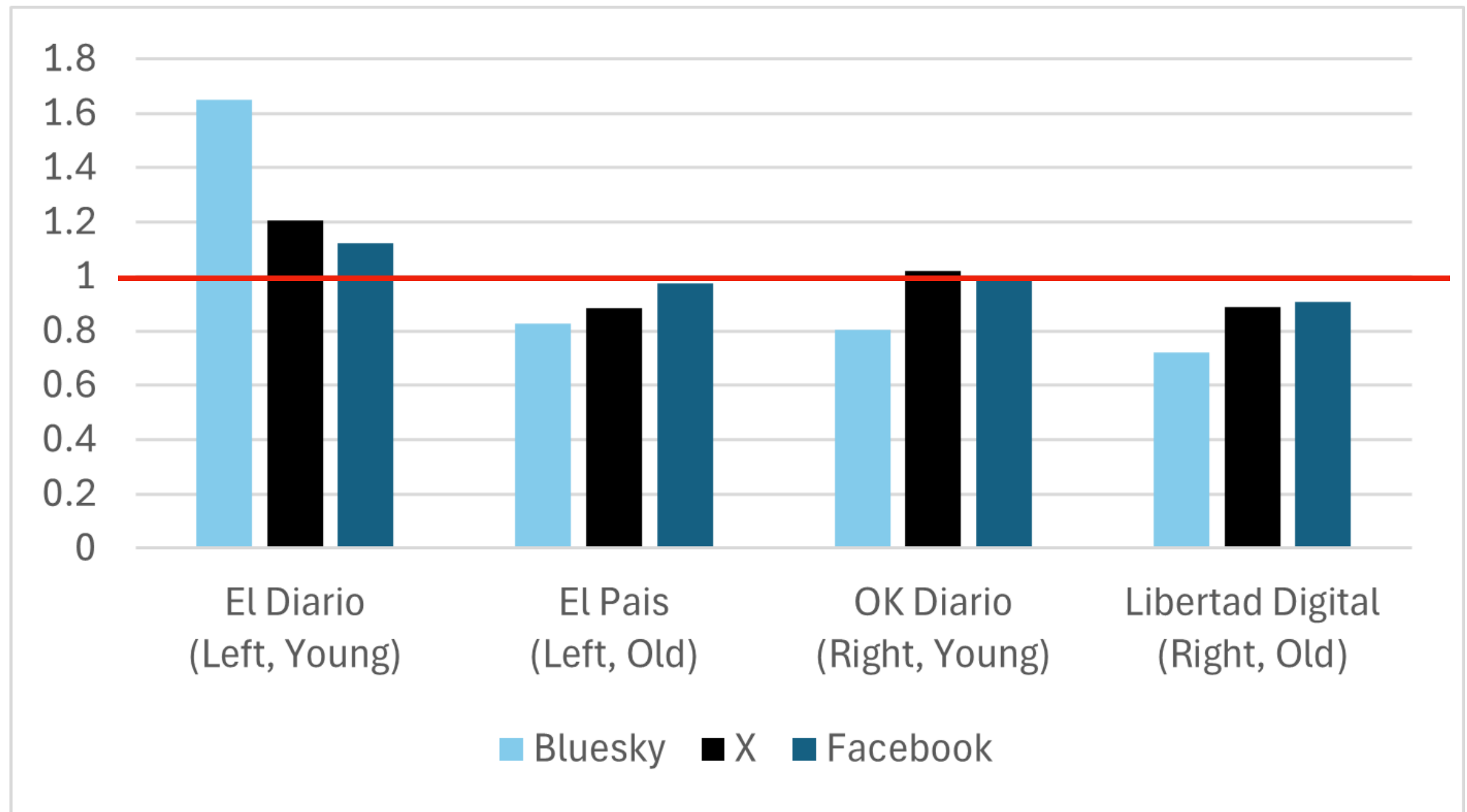
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* For a set of large newspapers in Spain and France

Assumption: Newspaper ideology = User ideology

Example of the data

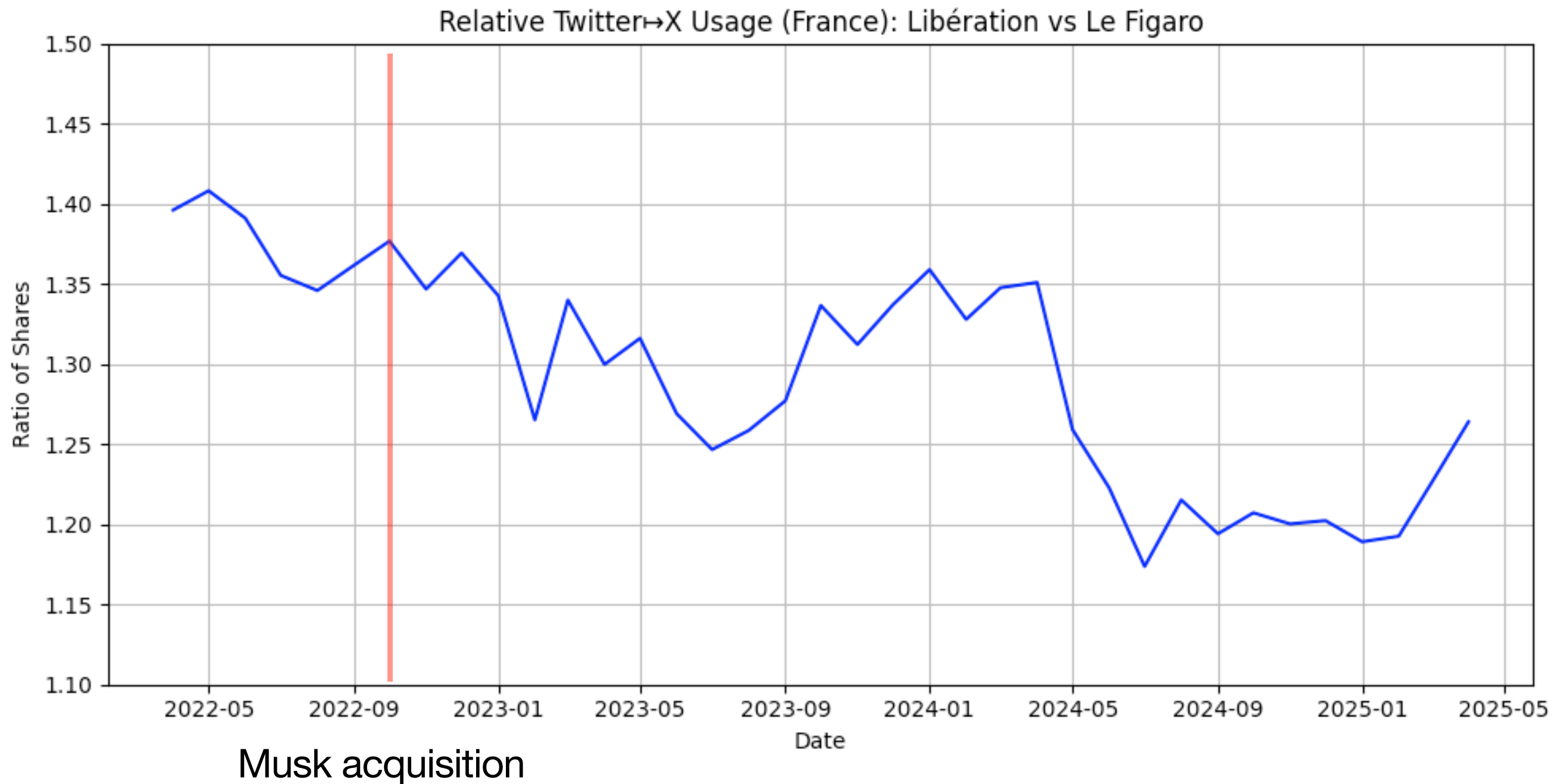
$$\text{Relative Propensity to Be on platform } j = \frac{\text{Proportion of readers of newspaper } i \text{ on platform } j}{\text{Average proportion of users on platform } j \text{ (across newspapers)}}$$



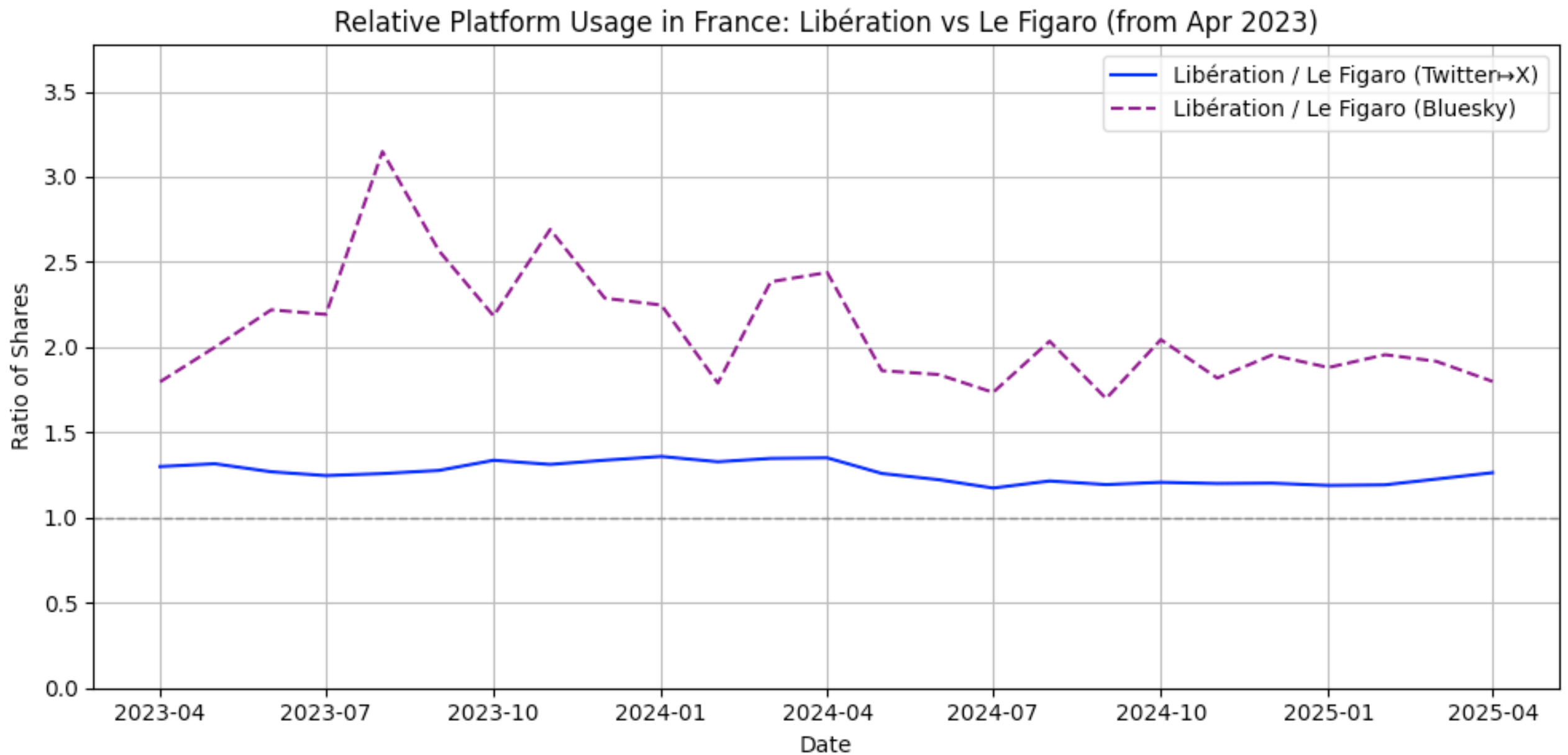
Average for the period 04/2022 - now

Example of the data

$$\frac{\text{Share of Liberation readers using X/Twitter}}{\text{Share of Le Figaro readers using X/Twitter}}$$



Example of the data



Demand model

$i = 1, \dots, N$ users on the internet

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Each user i has a type / ideology $\theta = 1, \dots, \Theta$

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$$U_{i,\theta,j,t} = \beta_0 + \sum_{\theta' \in \Theta} b_{\theta,\theta'} \gamma_{\theta',j} N_{\theta',j,t-1} + \xi_{\theta,j,t} + \epsilon_{i,j,t}$$

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“Prominency” of each type of content (algorithm)

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“Prominency” of each type of content (algorithm)

Demand model (2)

Berry's inversion. I can derive

$$\log \left(\frac{s_{\theta,j,t}}{s_{\theta,0,t}} \right) = \beta_0 + \sum_{\theta'} b_{\theta,\theta'} \gamma_{\theta',j} N_{\theta',j,t-1} + \xi_{\theta,j,t}$$

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Remark: each type is a market (connected to the other markets)

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I first estimate $\lambda_{\theta,\theta',j}$'s (OLS) and then **recover** $b_{\theta,\theta'}$'s and $\gamma_{\theta',j}$'s

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What is $N_{\theta,j,t-1}$? Number of users of newspaper / ideology θ in using j

Identification

Viewer to creator
Constant across platforms

$$\lambda_{\theta, \theta', j} = b_{\theta, \theta'} \gamma_{\theta', j}$$

Viewer to creator, per platform

Creator specific
Per platform

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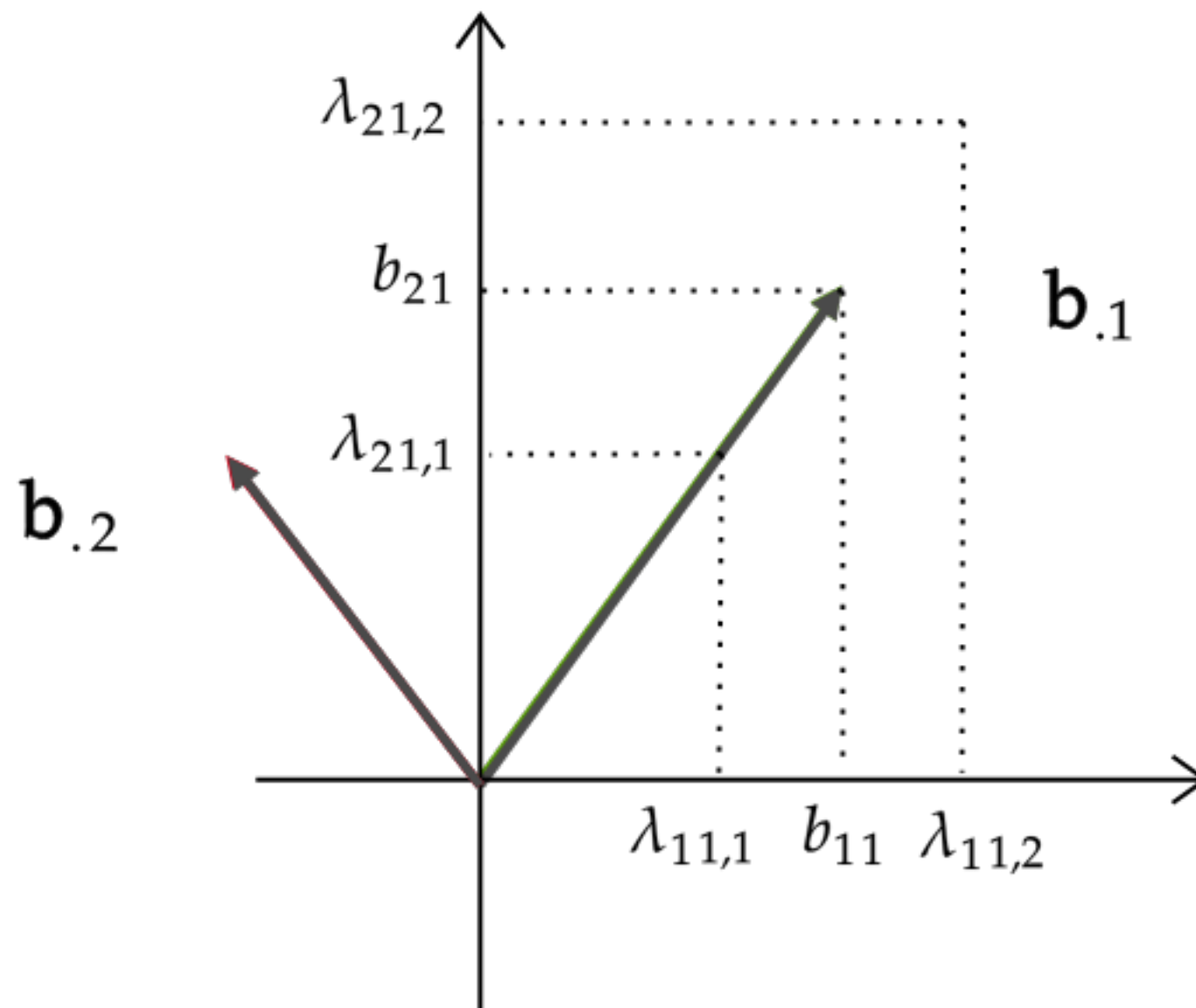
Creator specific
Per platform

If right wing users respond relatively more to the existence of right wing users in platform A than in platform B, then

Promotion of the right in A > Promotion of the right in B

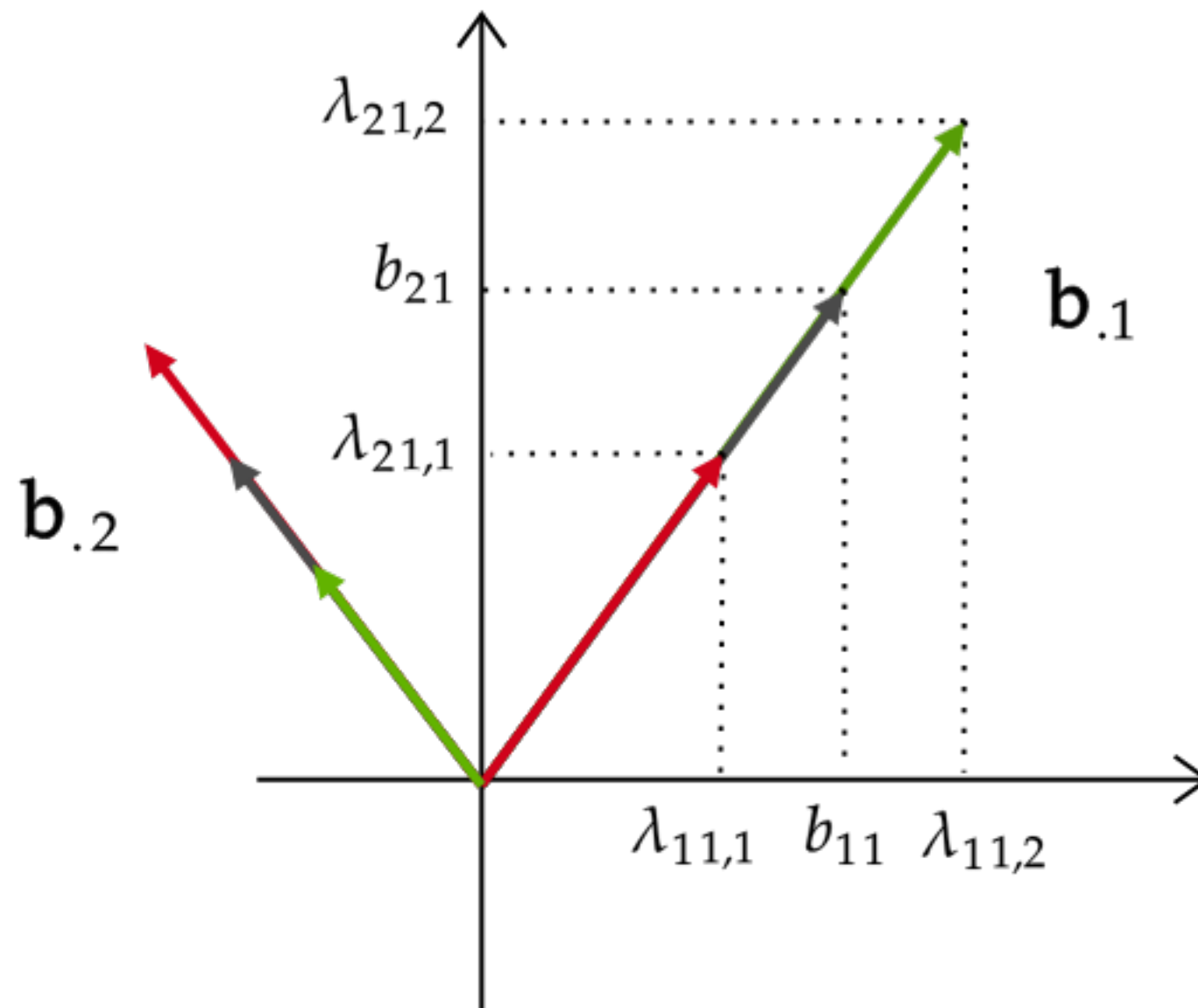
Is it identified? (2 platforms, 2 types)

Yes! Under the constraint that $\sum_{\theta=1}^{\Theta} \gamma_{\theta,j} = 1$ $\gamma_{\theta,j} \geq 0$



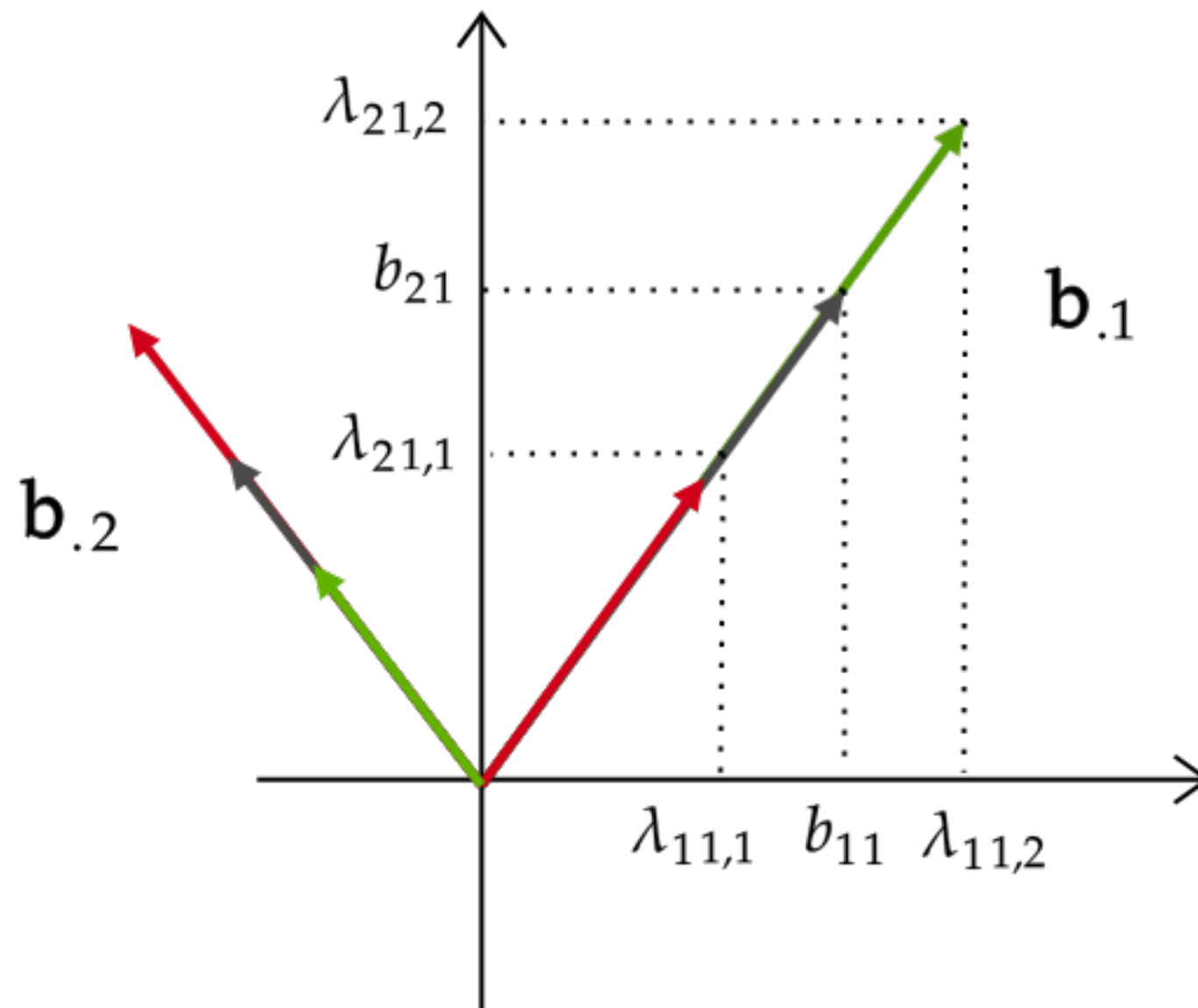
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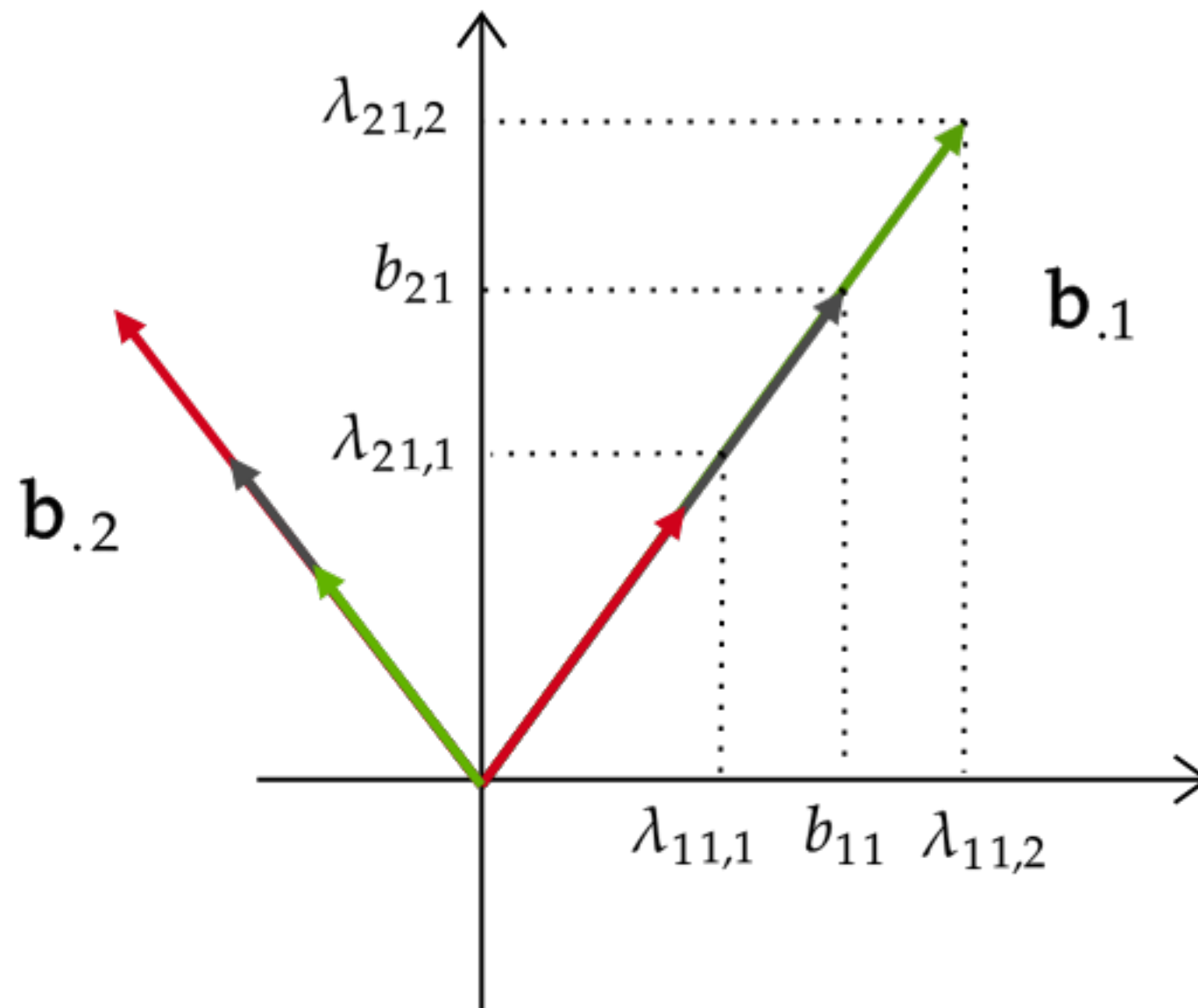
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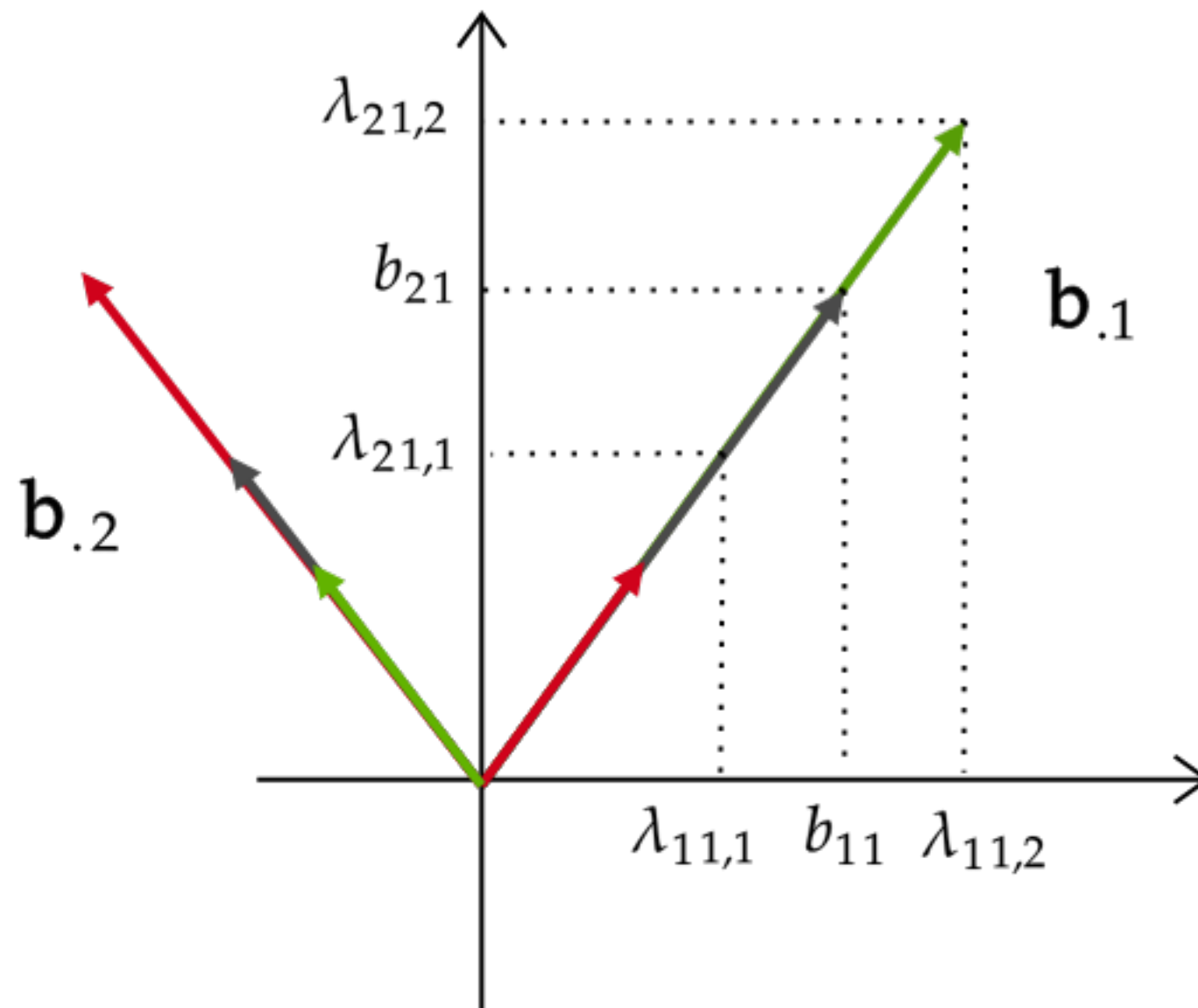
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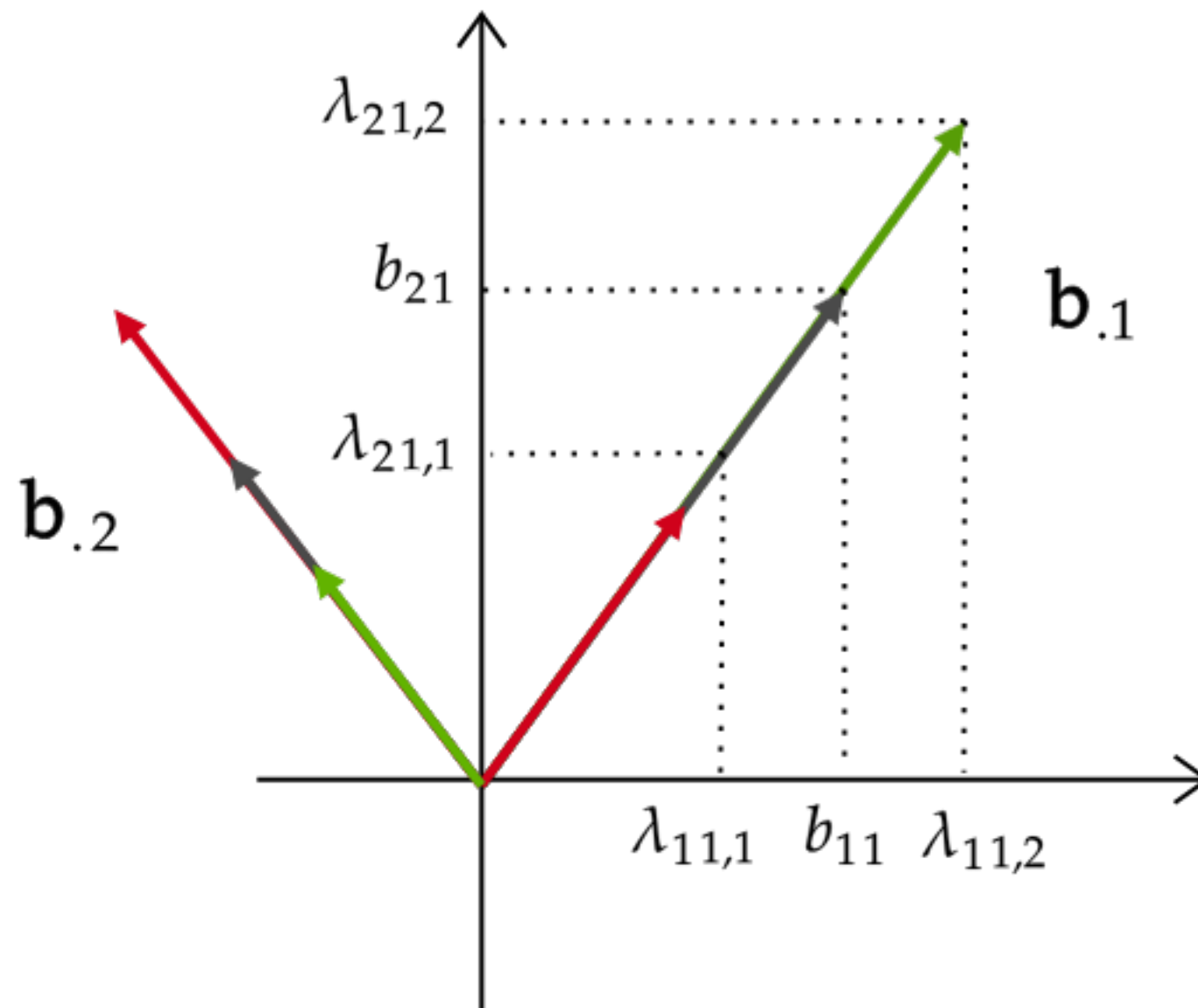
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Yes! Under the constraint that $\sum_{\theta=1}^{\Theta} \gamma_{\theta,j} = 1$ $\gamma_{\theta,j} \geq 0$



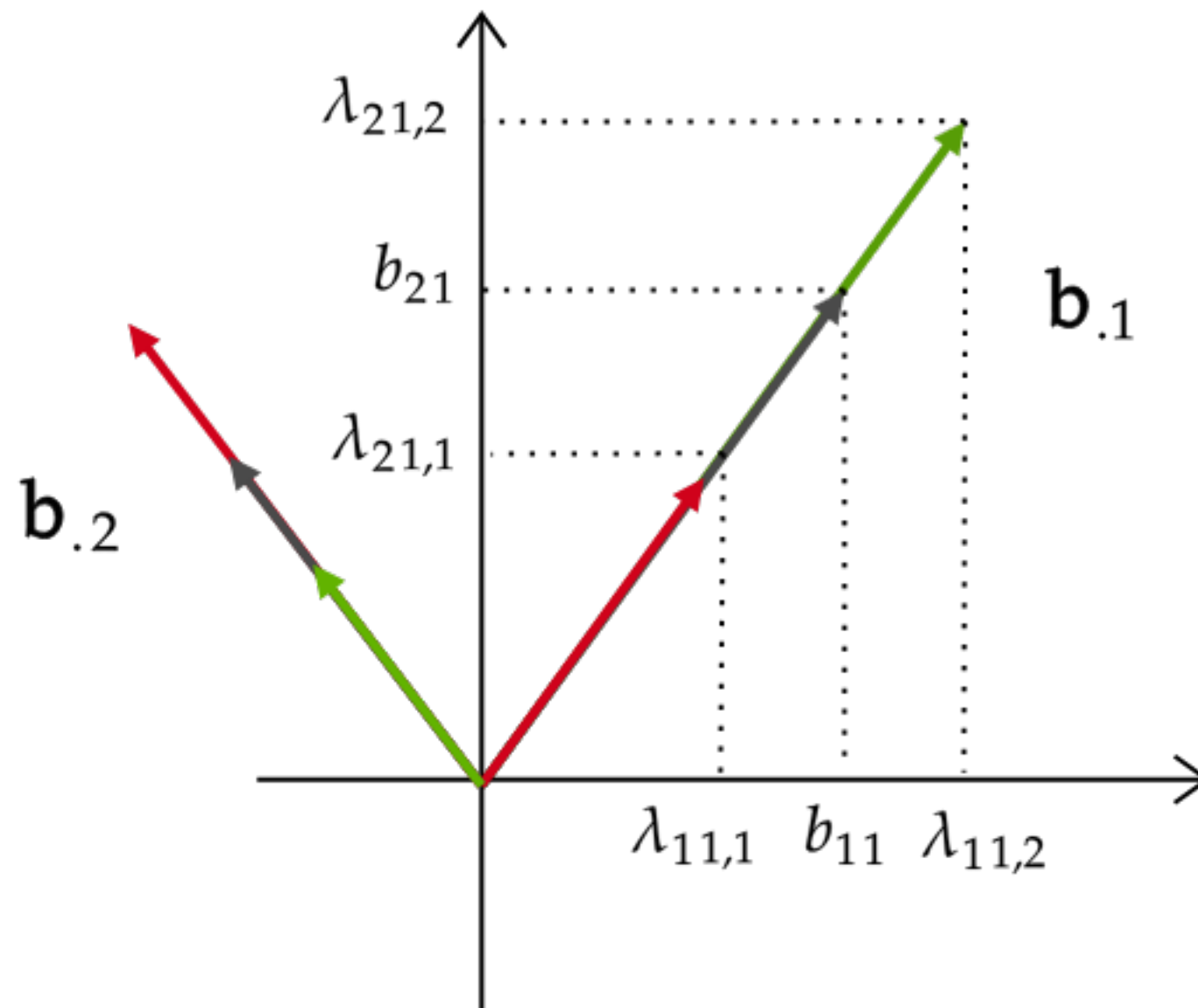
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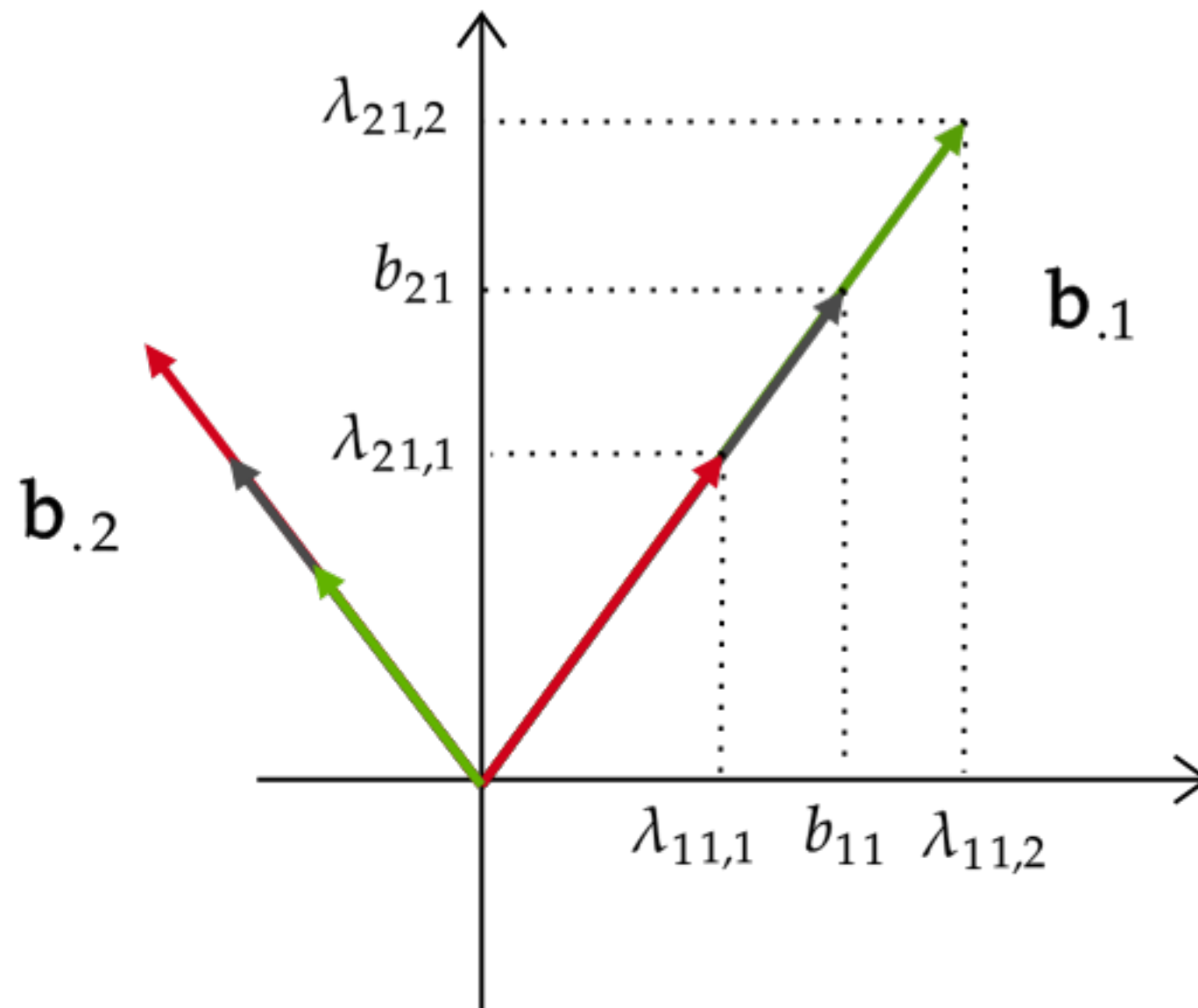
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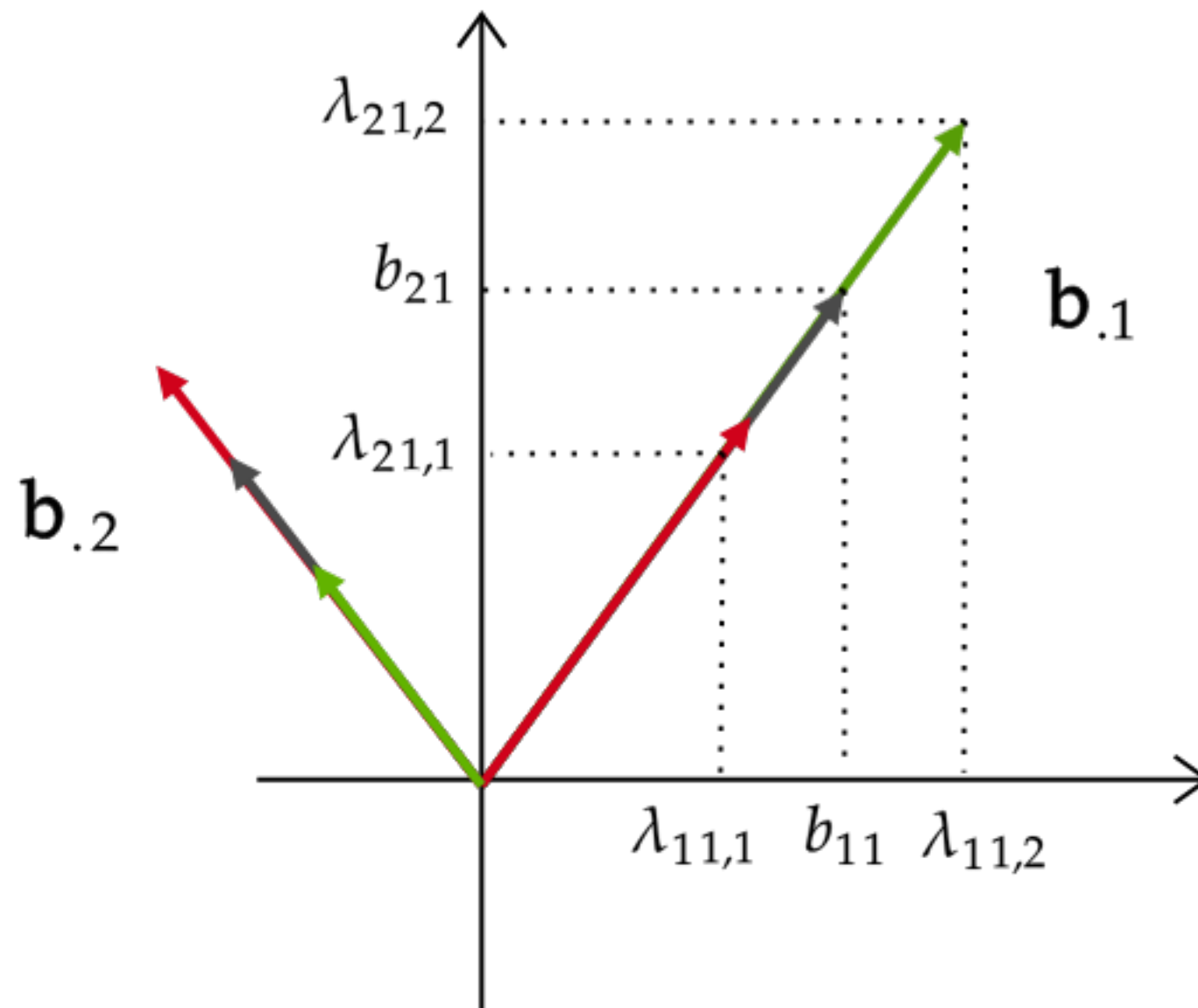
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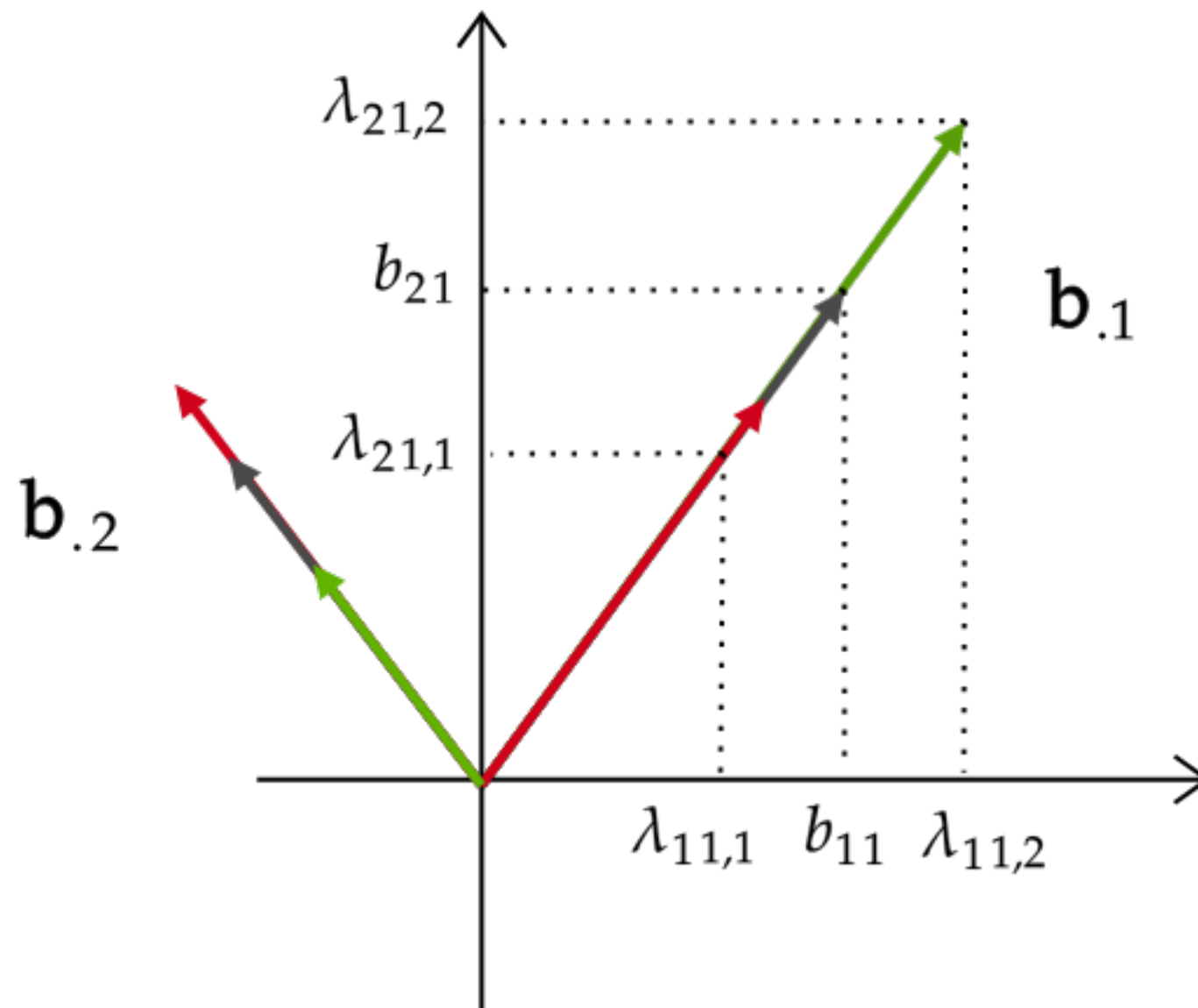
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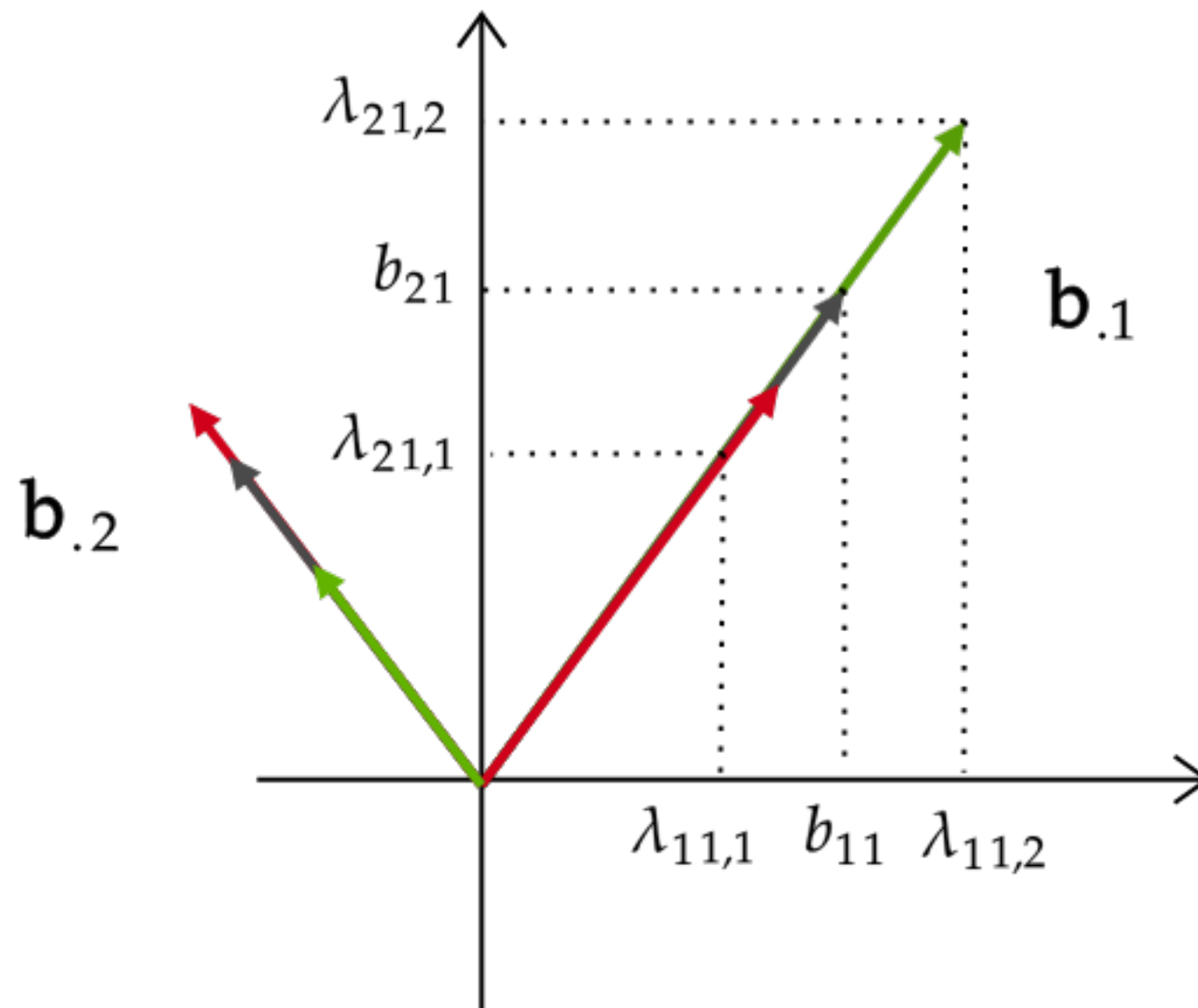
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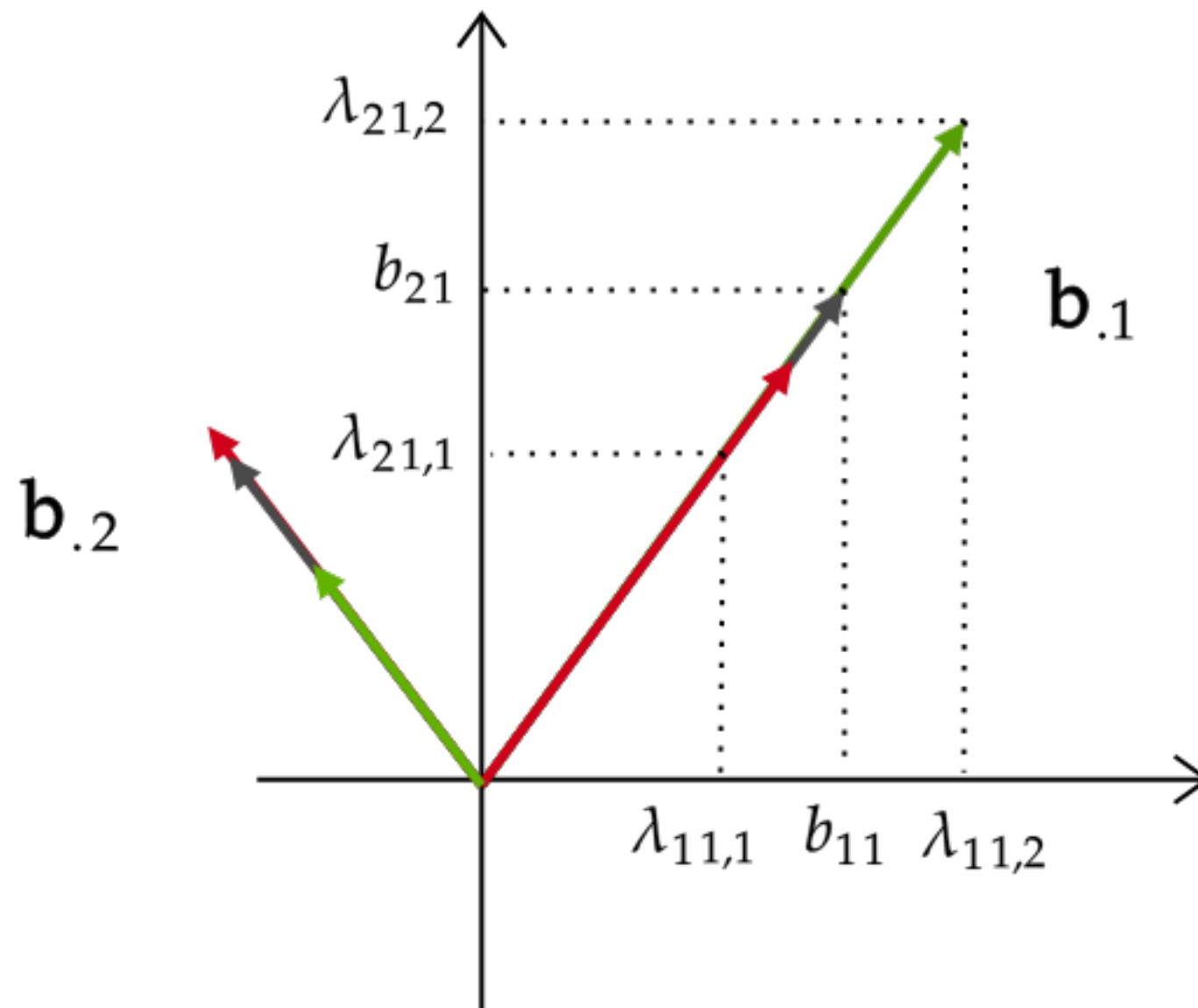
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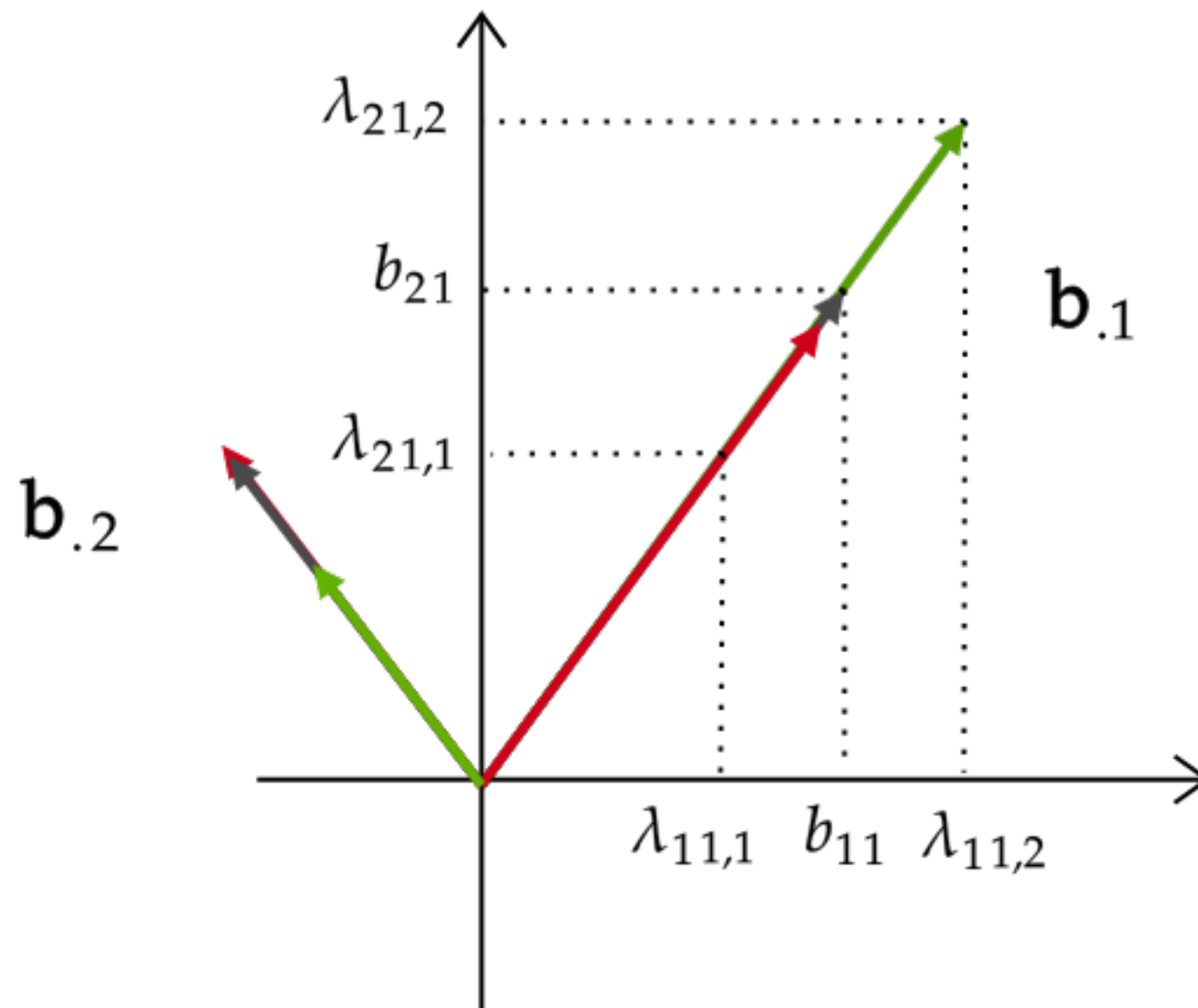
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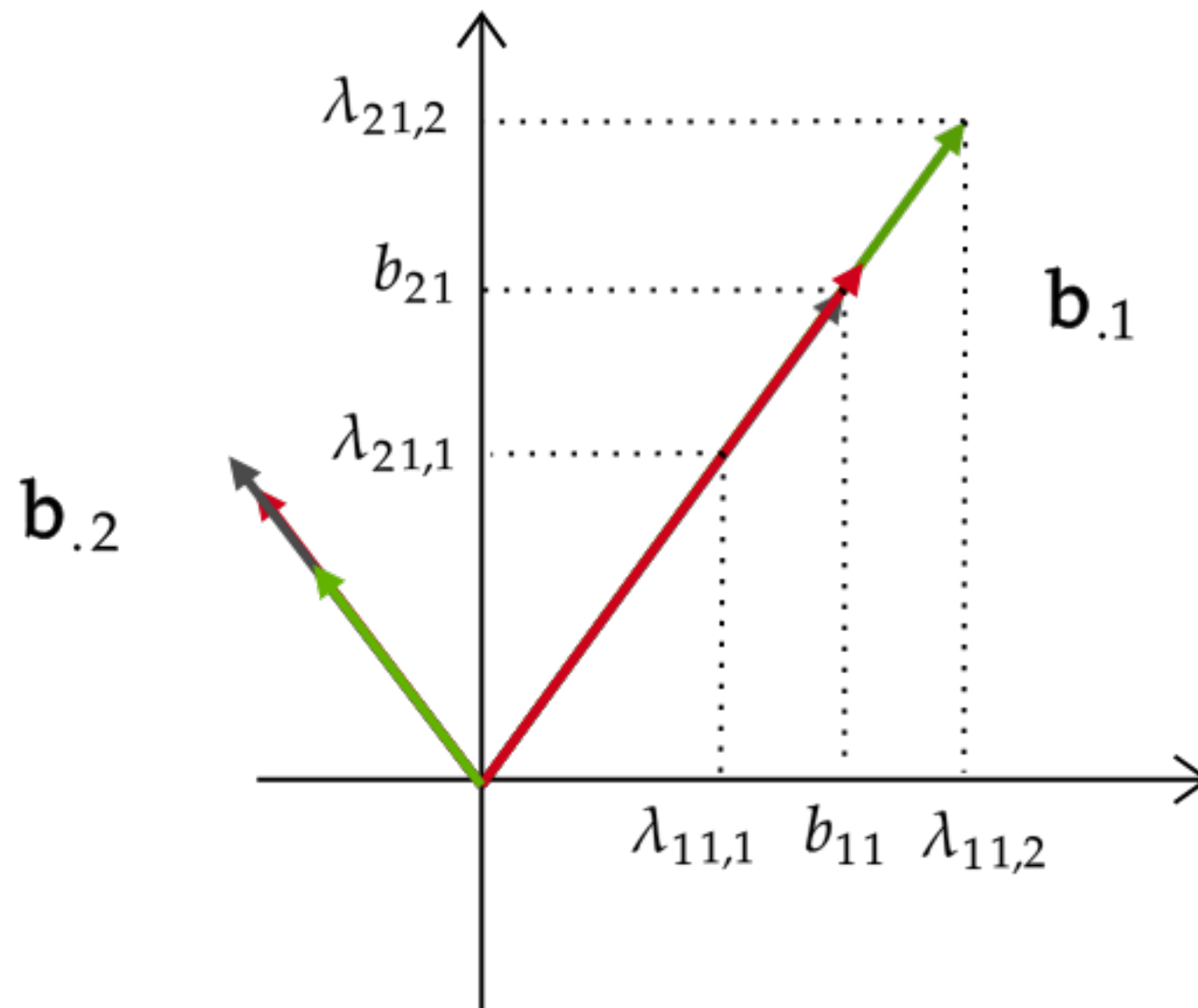
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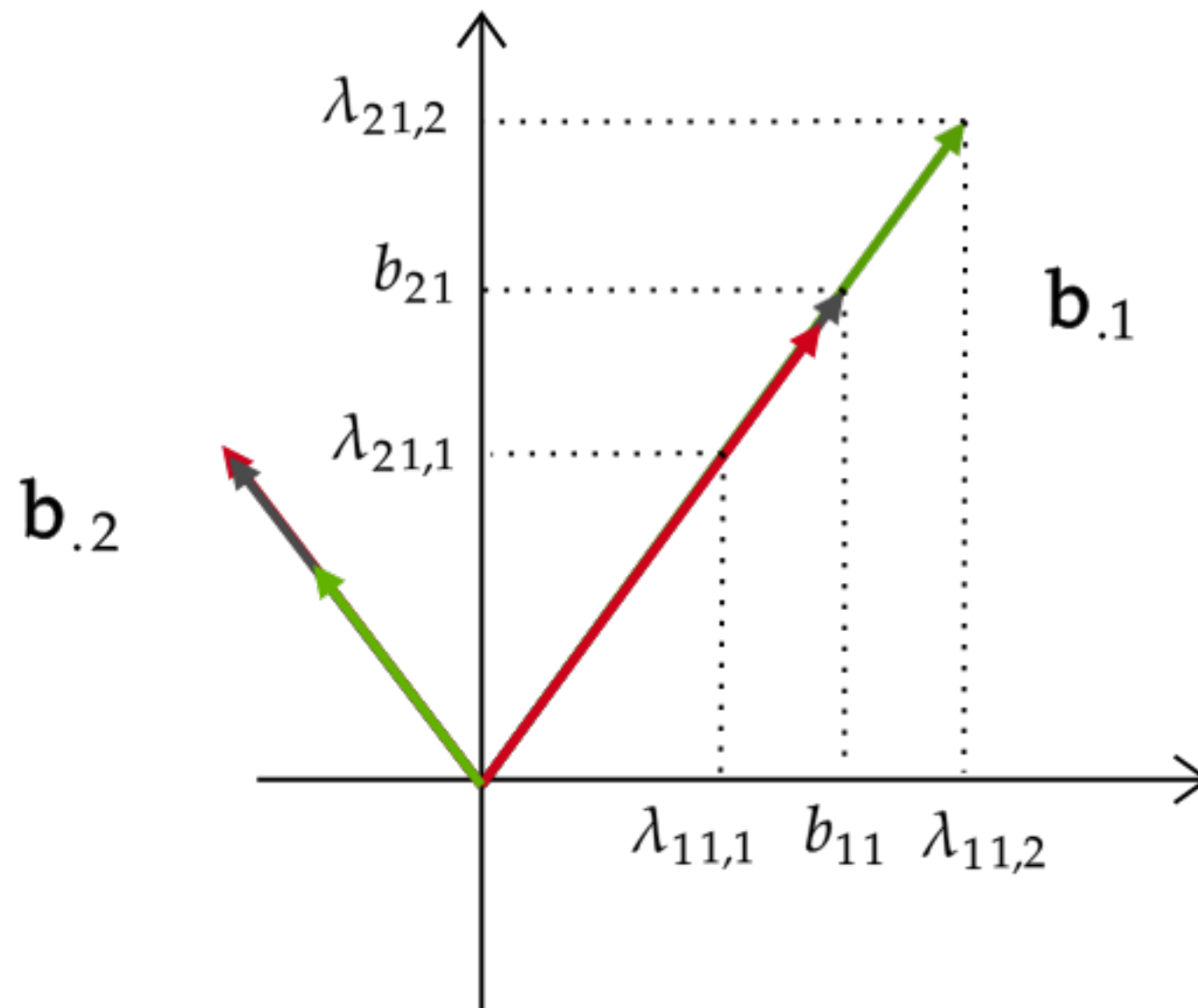
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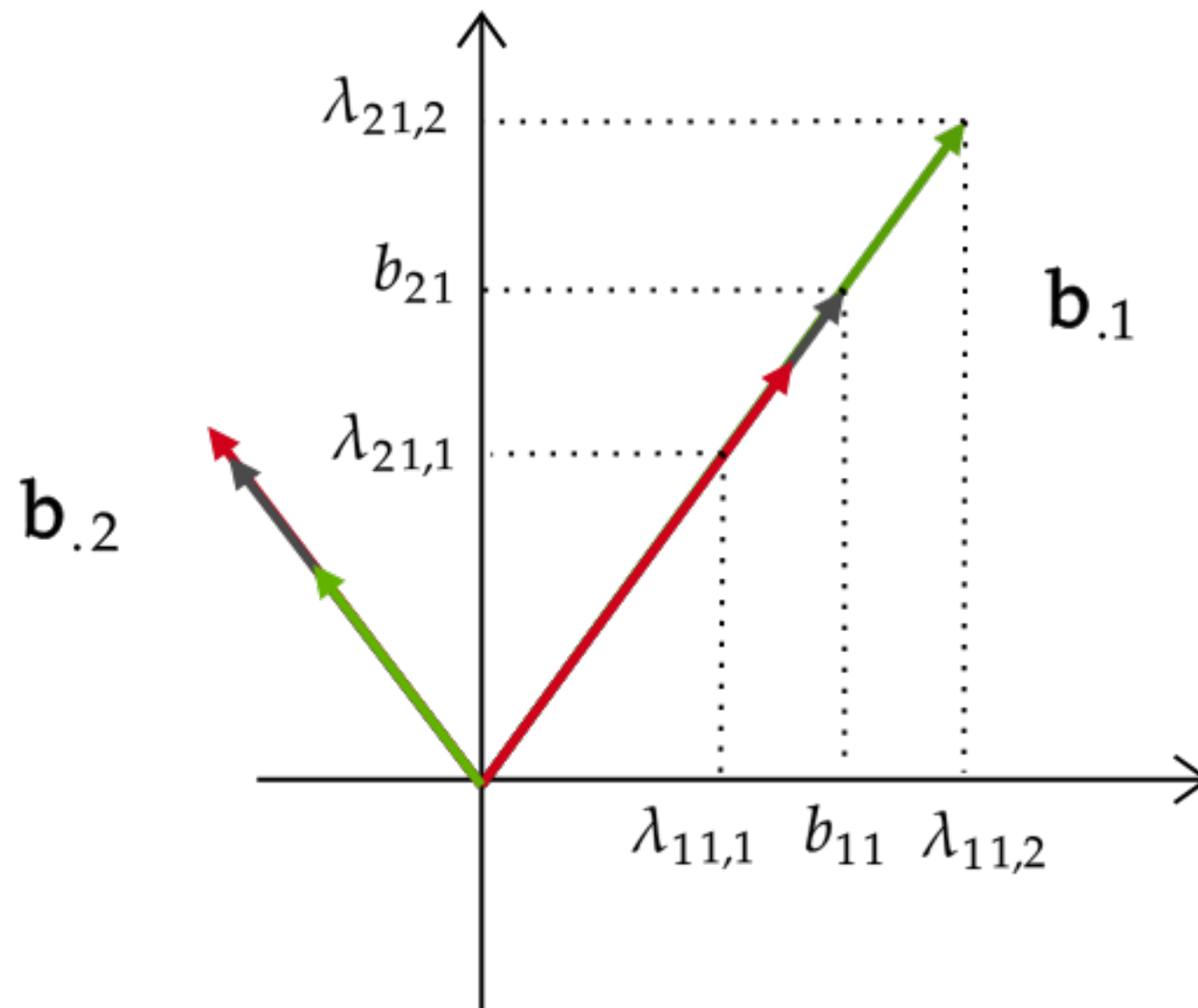
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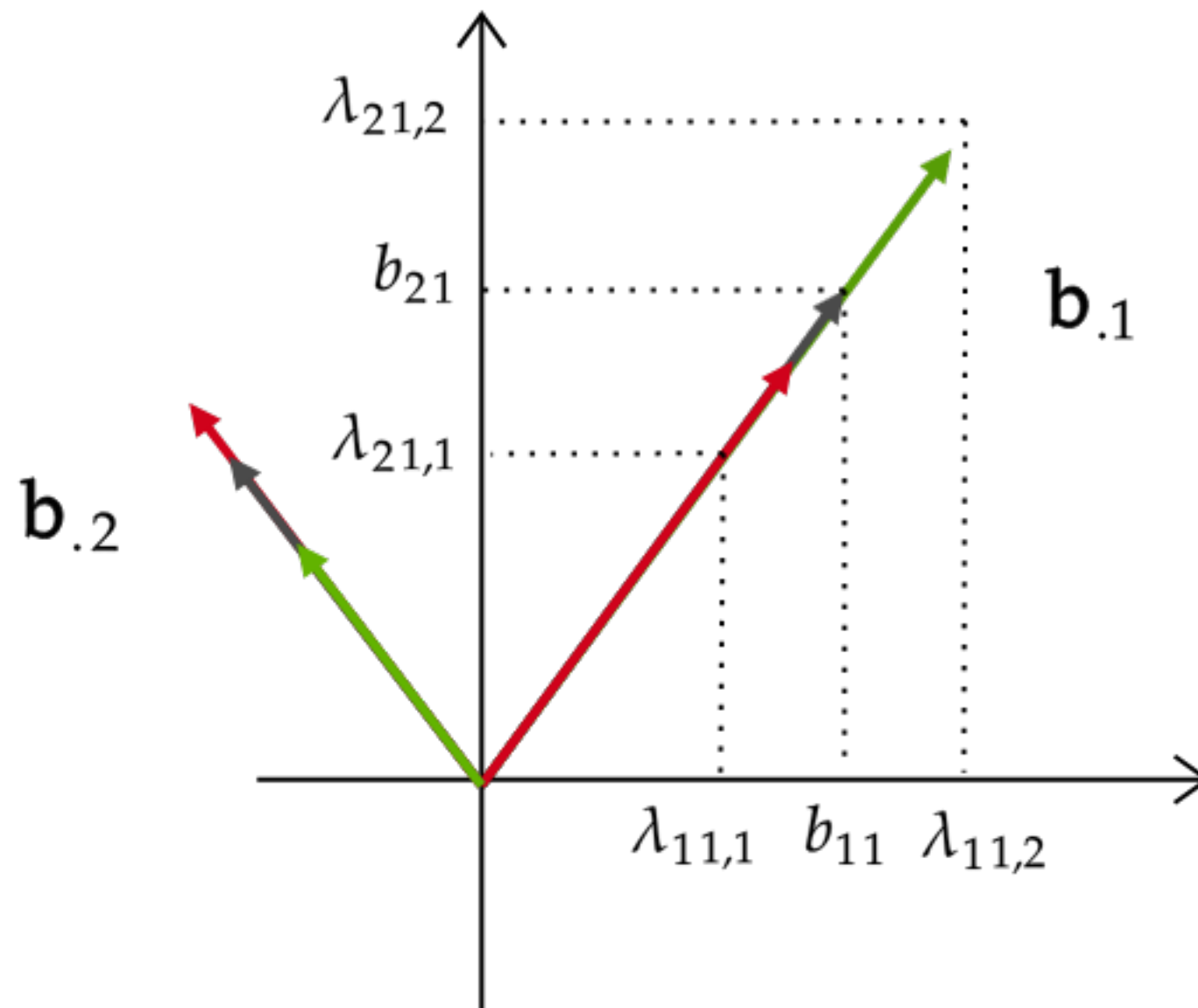
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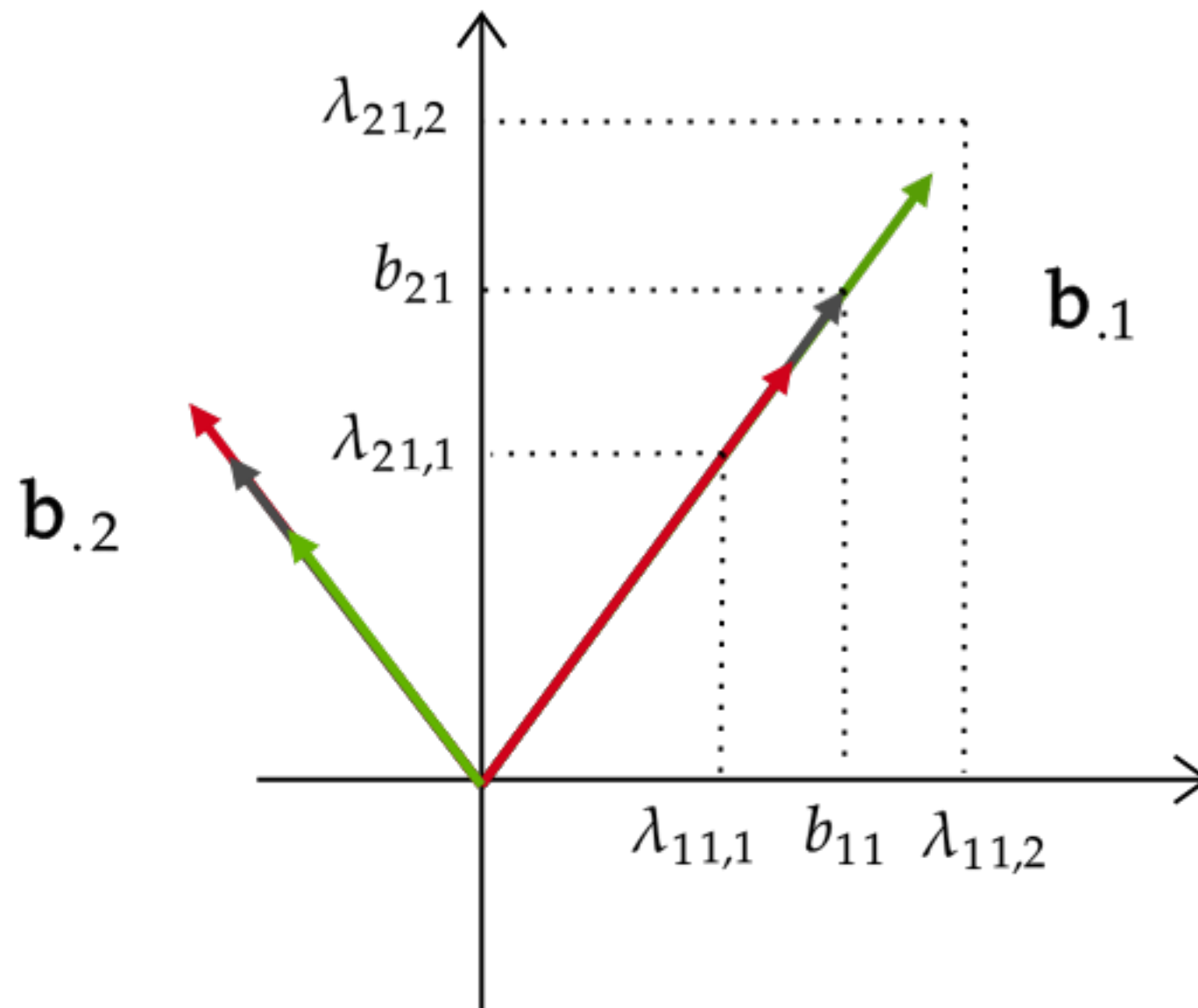
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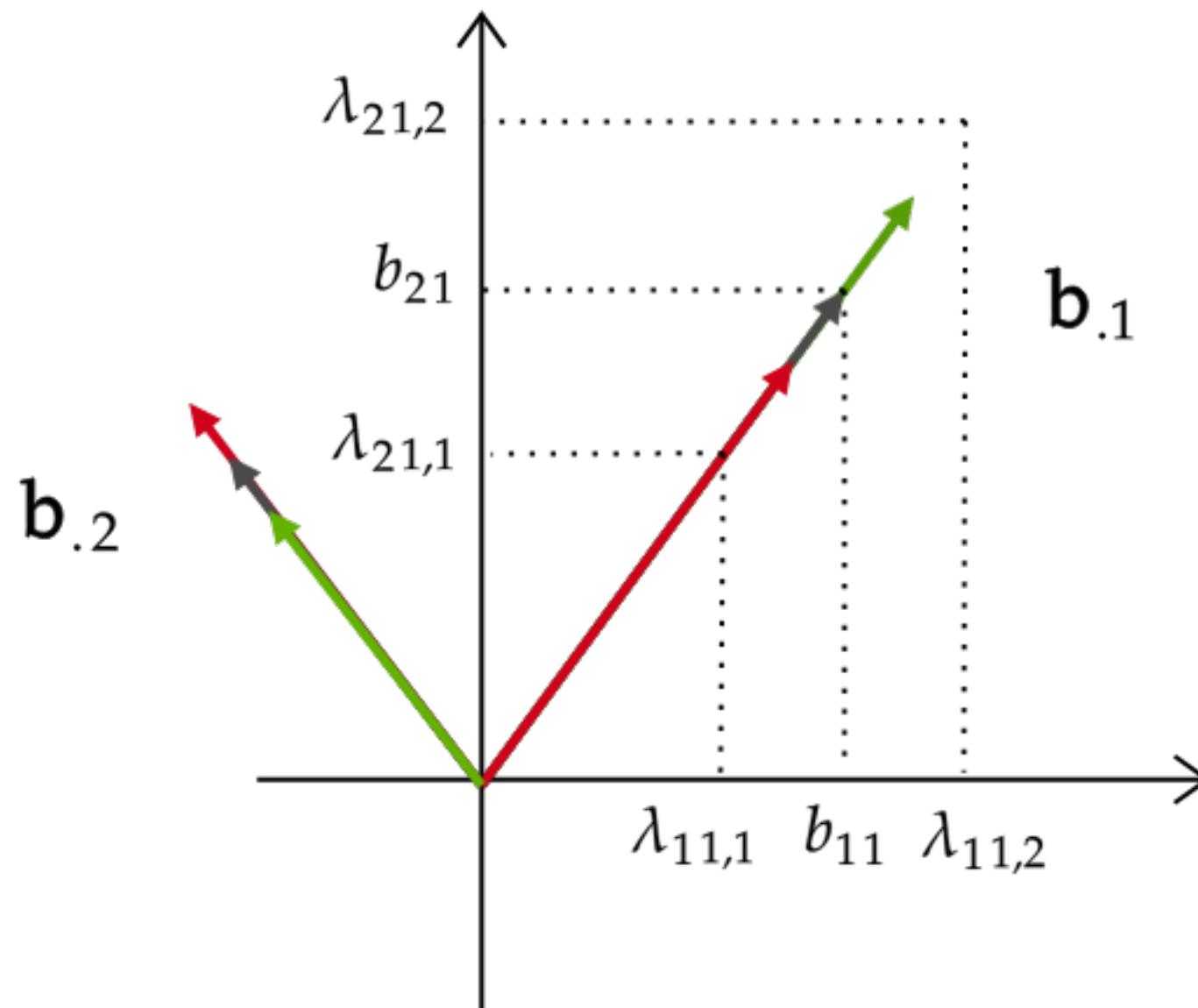
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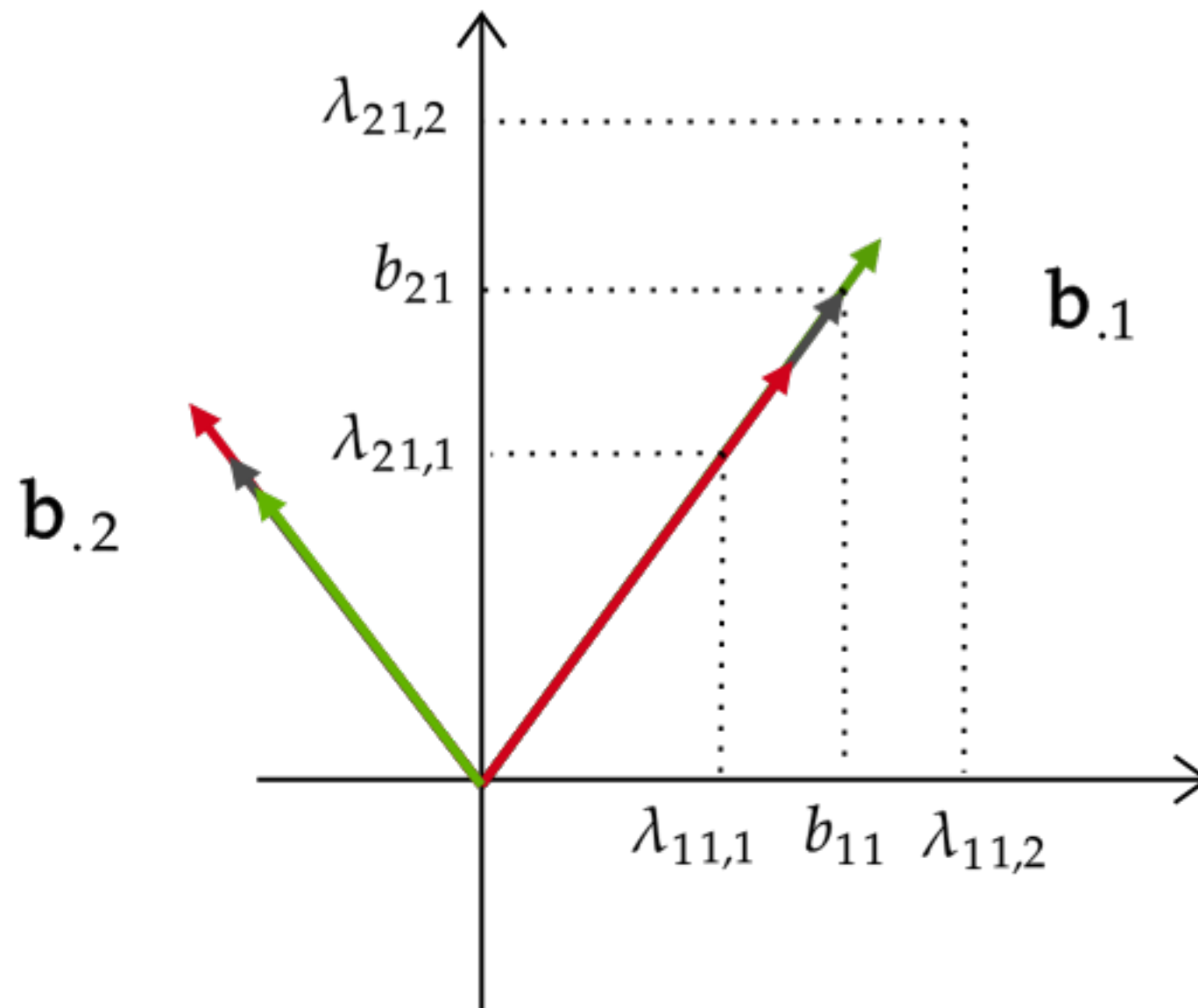
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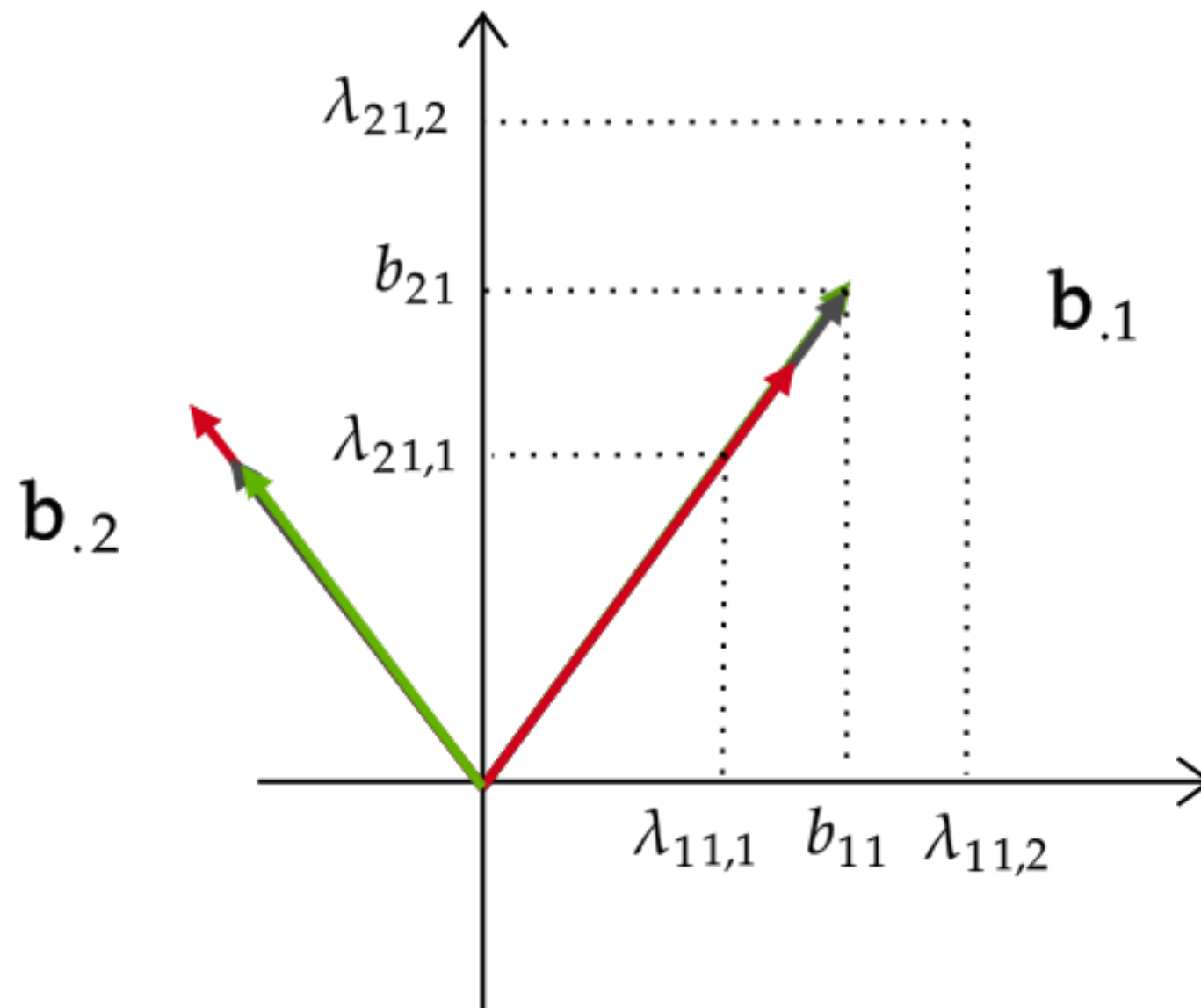
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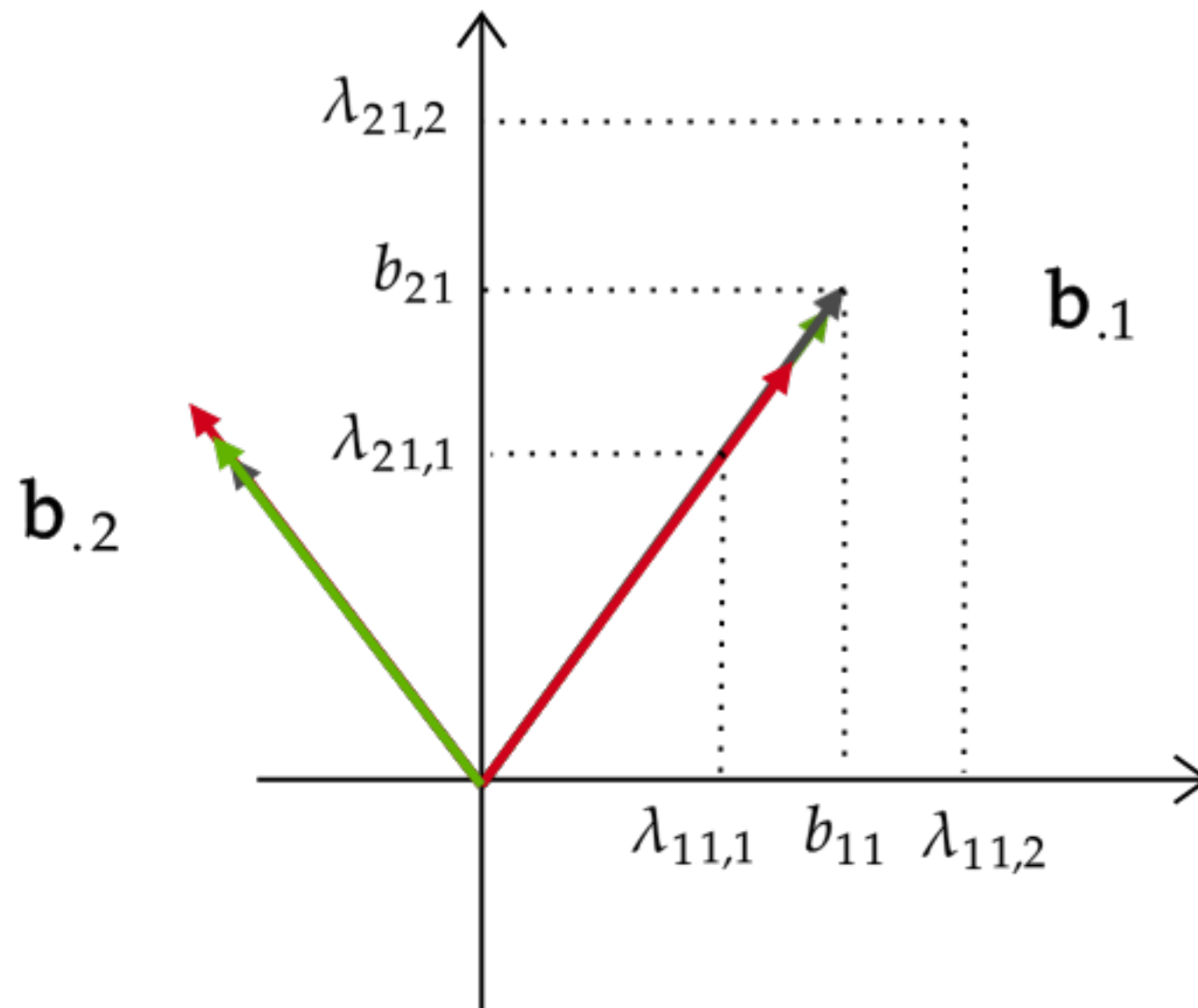
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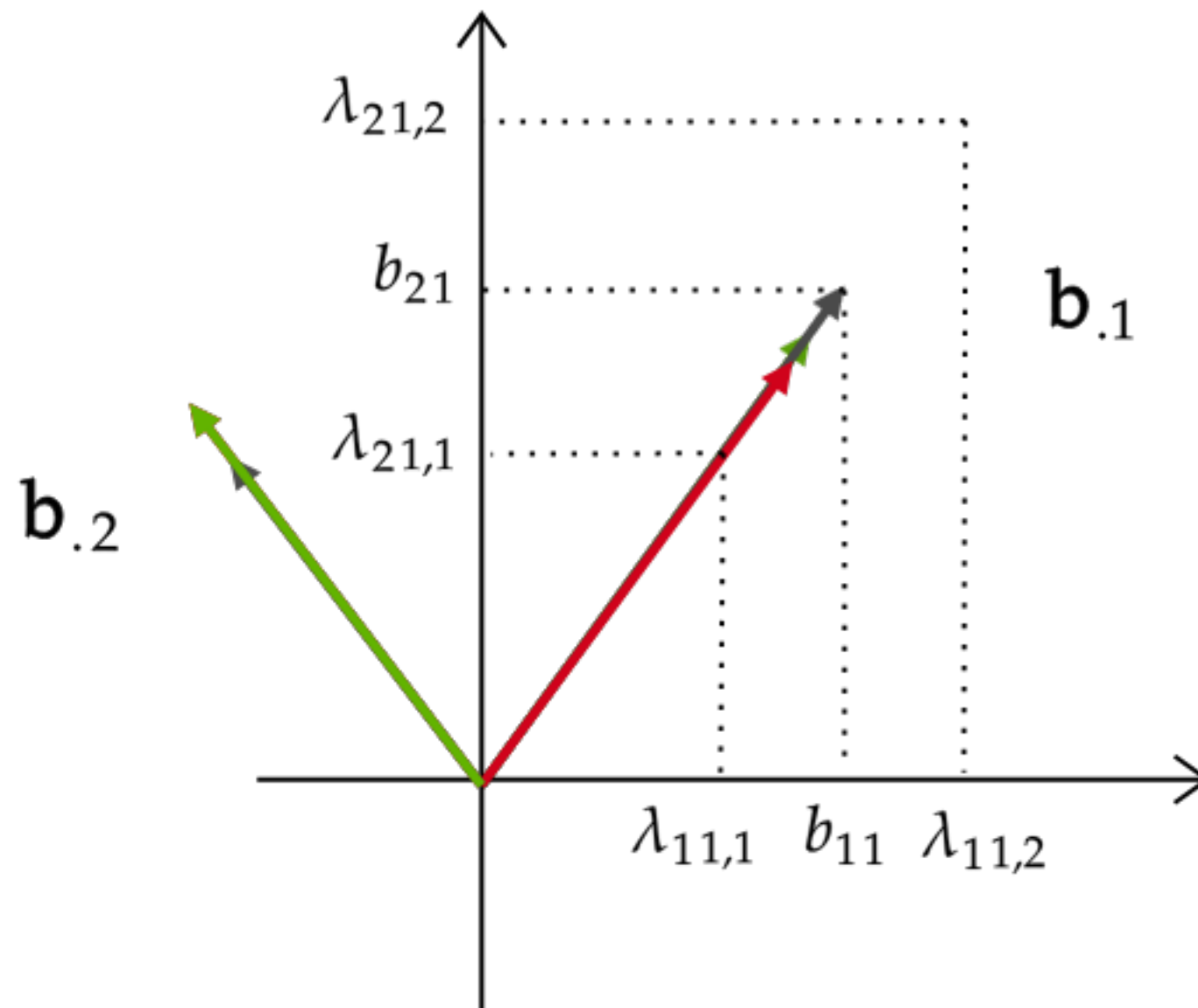
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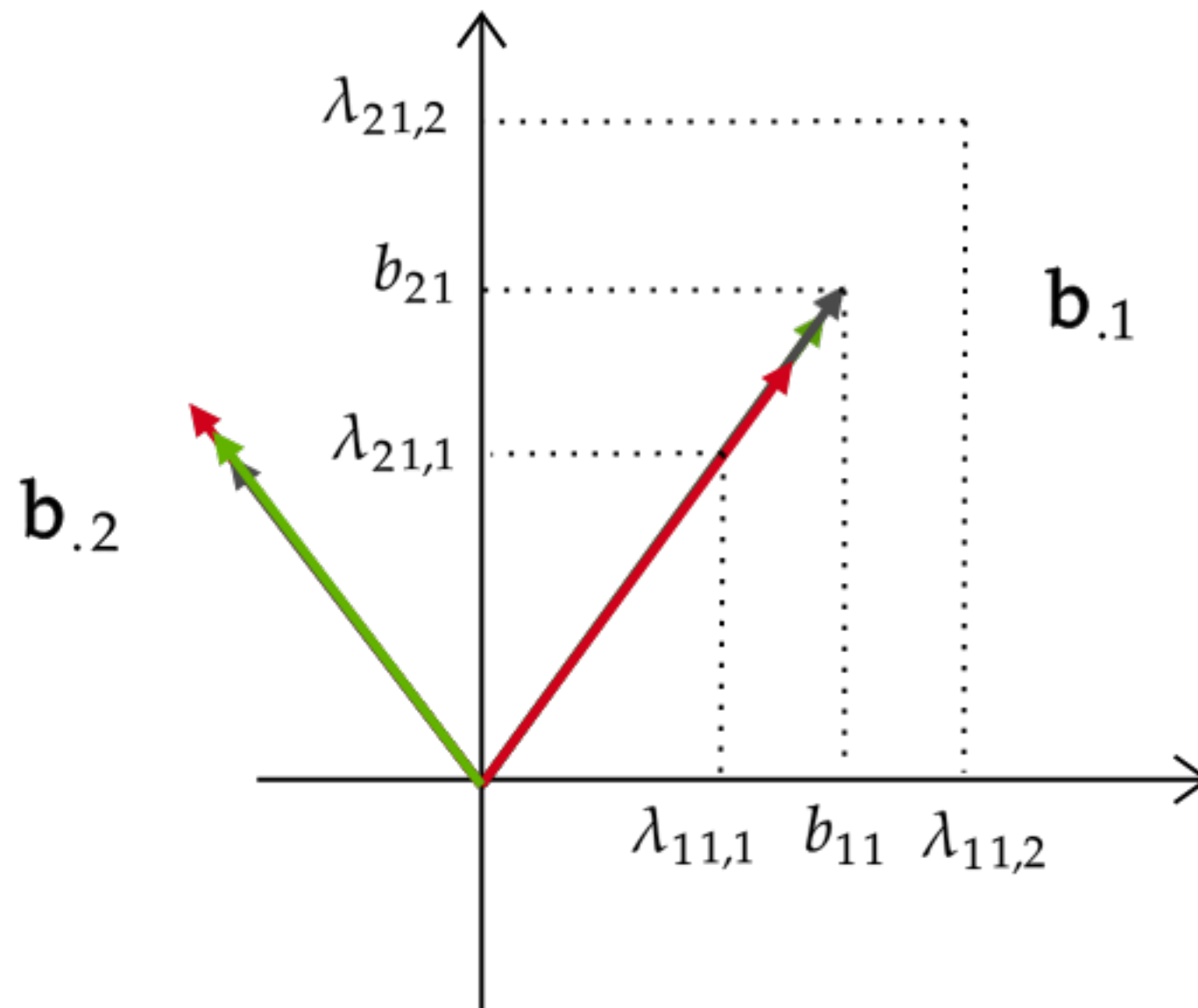
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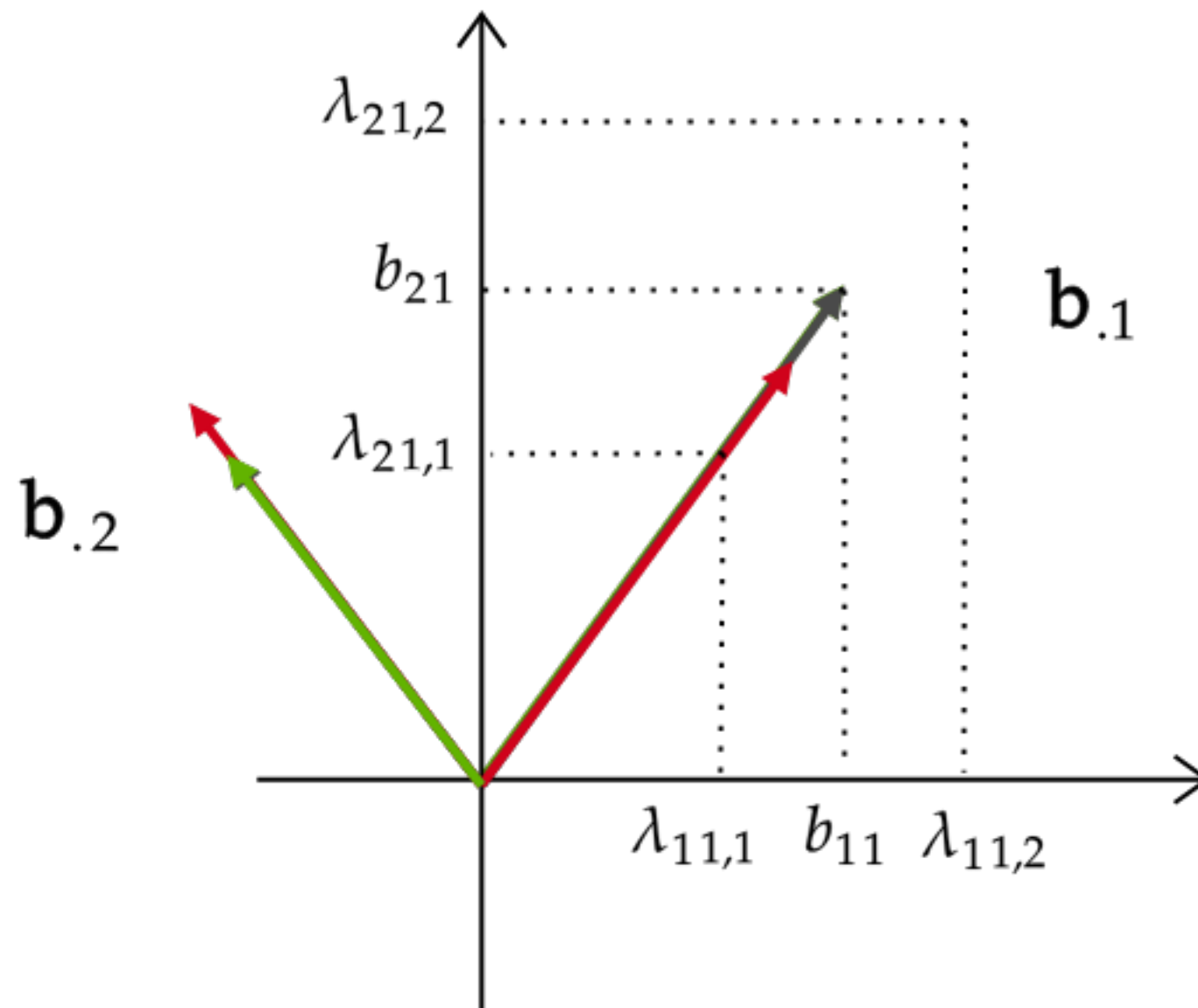
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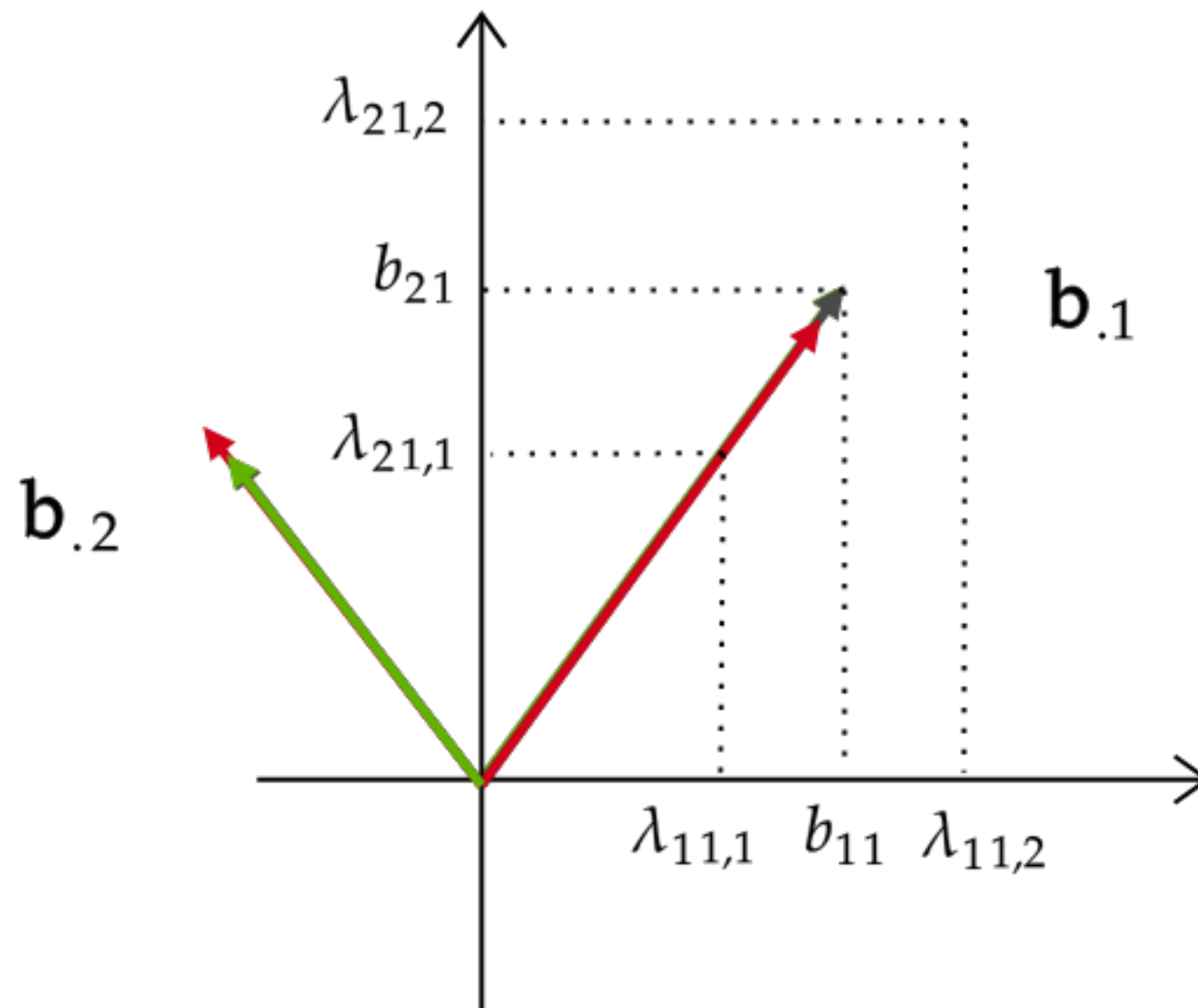
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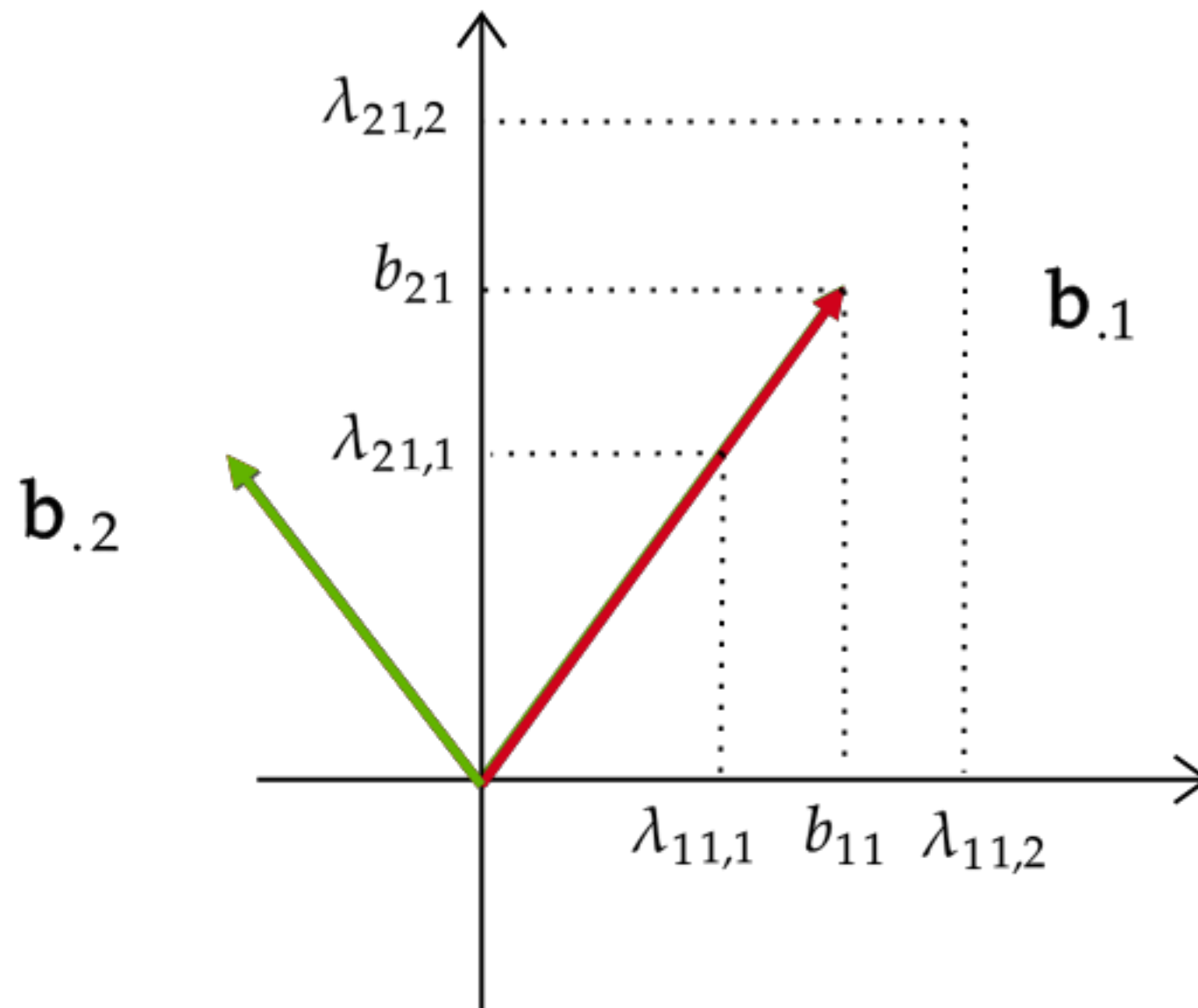
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Is it identified?

If more types than platforms $\rightarrow$ over-identification

+ Need to account for the noise from the estimation of λ

Recap

I have a sketch of a model

I have “the” data

Recap

I have a sketch of a model

I have “the” data

Things to consider

Endogeneity: using France as an IV to estimate Spain?

Users changing ideology / newspaper due to X?

Recap

I have a sketch of a model

I have “the” data

Things to consider

Endogeneity: using France as an IV to estimate Spain?

Users changing ideology / newspaper due to X ?

Overcome / minimize the reflection problem

Find an (optimal) way to estimate B and γ 's

Allowing for multihoming (already in the literature)